PARIKSHA365 — STUDY NOTES

Reasoning — for Banks

> Banking focus: puzzles (linear, circular, floors, scheduling), input-output, syllogism (advanced), coded inequality, data sufficiency, machine I/O

BANKS

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

⚠ EXAM ALERT

∗∗Examiner Blueprint — Reasoning** ∗∗Top 8 most-tested topics (question share across SSC + Banks):** | Rank | Topic | SSC CGL T-1 | IBPS PO Pre | IBPS PO Mains | |---|---|---|---|---| | 1 | Puzzles & Seating Arrangement | 3–5 | 3–5 | 15–20 | | 2 | Syllogism | 3–5 | 3–5 | 3–5 | | 3 | Inequalities / Coded Inequalities | 2–3 | 3–5 | 3–5 | | 4 | Number & Letter Series | 3–5 | 3–5 | 2–3 | | 5 | Blood Relations | 2–3 | 2–3 | 3–5 | | 6 | Direction & Distance | 2–3 | 2–3 | 2–3 | | 7 | Coding-Decoding | 3–5 | 2–3 | 2–3 | | 8 | Input-Output | 0–1 | 0–1 | 4–6 |

✔ KEY FACT

∗∗Study Priority Order** (return-on-study-time, highest first) 1. ∗∗Syllogism** — 2 focused hours → 3–5 nearly free marks in every paper. 2. ∗∗Inequalities** — 1 hour coded + 1 hour direct → 3–5 marks in every Banks paper. 3. ∗∗Series** — 3 hours drilling squares/cubes/primes + pattern types → 3–5 marks in every prelim. 4. ∗∗Blood Relations** — 2 hours; draw a family tree for every question and you will never go wrong. 5. ∗∗Direction & Distance** — 2 hours; coordinate method (X-Y axes) turns every question mechanical. 6. ∗∗Coding-Decoding** — 3 hours; always test the rule on both given examples before applying. 7. ∗∗Seating & Linear Arrangement** — 8–10 hours; the single topic that decides whether you clear prelims comfortably. 8. ∗∗Floor / Box / Calendar Puzzles** — 6–8 hours; mandatory for Banks Mains selection.

How to use this book

Reasoning is not a single topic — it is a **cluster of pattern families**. Once you can see the pattern in 5 seconds, the rest is bookkeeping.

The same type-first discipline applies throughout. For every topic:

- **Opener** — why it appears, what you must memorise cold, universal attack plan.
- **Types** — a focused set. Many topics settle around 8-15 types; complex topics (Puzzles, Input-Output, Non-verbal) may need 20-25. **There is no upper cap** — a type is included if it unlocks a distinct pattern that will earn marks; never padded just for a round number.
- **For each type** — recognition cue → core insight → shortcut-first solution (with derivation) → 2-3 variations → self-check.

Diagram accuracy — a hard rule

Reasoning questions whose answer depends on a visual (mirror-image, water-image, dice, paper-fold, cube-counting, embedded figures, mirror-Line-of-symmetry, figure-completion) can only be taught with **precise, to-scale illustrations** — not rough ASCII or hand-wave sketches. Wrong visuals confuse worse than no visuals.

- This text book teaches the **method** for each visual type (how a mirror flips about a vertical line, how dice opposite-face rule works, how a folded paper unfolds, how to rotate a 3-D cube in your head).
- Figures are exact SVG, physically verified before publication. No figure is shipped until a human has confirmed the mirror / rotation / unfold is physically correct.
- If a specific question cannot be diagrammed faithfully, it is deferred — never replaced with a crude approximation.

PART 1 — SERIES (NUMBER, LETTER, ALPHANUMERIC)

1.0 Opener

Why: 2-4 questions every prelims. Cheapest marks if you drill the patterns.

Mental-math arsenal: - Squares 1-30, cubes 1-15 (for number series). - Alphabet positions (A=1, N=14, Z=26) and reverse positions (A=26, Z=1). - Prime list up to 50.

Attack plan: compute first differences. If constant → AP. If not → second differences or ratio or known sequence.

Series type recognition table:

What you see in Δ^1	Pattern	Type
Constant	AP	Arithmetic
Constant ratio	GP	Geometric
1, 4, 9, 16 ...	Square series	n^2
1, 8, 27, 64 ...	Cube series	n^3
Δ^1 itself is AP	Quadratic	2nd diff constant
Two interleaved APs	Alternate series	Split odd/even
×2+1, ×2+1	Mixed operations	Apply recipe
2, 3, 5, 7, 11 ...	Prime series	List of primes

✎ WORKED EXAMPLE

Quick example. Find missing: 3, 6, 11, 18, **?**, 36

```
**Δ¹:** 3, 5, 7, ?, ? **Δ²:** 2, 2, 2 → constant → next Δ¹ = 9 **Missing term:** 18 + 9 = **27** **Verify:** 27 + 9 = 36 ✓ (Δ¹ = 9, Δ² = 2 → next Δ¹ = 11; 27 + 9 ≠ 36 × next; wait: next diff after 27 is 36-27=9 ✓)
```

Types

1. **Arithmetic progression** — constant difference.
2. **Geometric progression** — constant ratio.
3. **Square / cube series** — n^2, n^2+1, n^2-1, $(n+1)^2$...
4. **Fibonacci-style** — each term = sum of two previous.

5. **Alternate series** — odd positions form one pattern, even positions another.
6. **Mixed operations** — ×2 +1, ×2 +1 ... (recurring recipe).
7. **Prime / composite based**.
8. **Letter series** — differences in alphabet positions; forward/backward/alternating.
9. **Alphanumeric** — letter + number pairs, each tracked separately.
10. **Missing term at position k** — recognise pattern, extrapolate.

For each type: **compute the Δ row and match to the recognition table above.**

PART 2 — CODING-DECODING

2.0 Opener

Memorised: - A=1…Z=26 (and Z=1…A=26 reverse). - Common offsets: +1 (B→A), +2, ×reverse.

Types

1. **Letter shift coding** — every letter shifted by k positions (CAT → DBU with k=+1).
2. **Reverse letter coding** — letter replaced by (27 − position) letter (A↔Z, B↔Y).
3. **Position-dependent coding** — even positions shifted differently from odd.
4. **Word coding by blocks** — split word, shift blocks.
5. **Substitution coding** — whole-word replacement (sun = tree, moon = flower…), questions require cross-sentence inference.
6. **Mathematical coding** — digits + symbols used as letters.
7. **Chinese-coding / symbol-coding** (new pattern) — given a mapping table, decode.
8. **Letter-digit hybrid codes** — each letter has position + offset rule.

Shortcut: **always write the alphabet row + index row on your rough sheet first**, then slide the shift.

PART 3 — DIRECTION SENSE

3.0 Opener

Memorised: the 8 directions on a compass rose; the **rotation of a direction under "left"/"right" turn** (90° CCW / 90° CW). Pythagorean triples (for shortest-distance questions).

Attack plan: draw a clear cross (N up, E right); mark each move to scale; use Pythagoras for diagonal distance.

Types

1. **Sequential movement** — plot each leg, sum vectors.
2. **Final displacement** — Pythagoras on N-S vs E-W net movements.
3. **Left/Right turns at each step** — maintain current heading; apply rotation.
4. **Starting direction unknown** — deduce from constraints later in the problem.
5. **Shadow direction / time of day** — morning shadows point W, evening E.
6. **Two persons meeting** — relative-position geometry.
7. **Rotation + reflection combo** — someone turns 135° clockwise; track new heading.

Shortcut: **always finish one person's path fully before starting the next; don't merge.**

PART 4 — BLOOD RELATIONS

4.0 Opener

Memorised: the family tree notation — square for male, circle for female, horizontal line for spouse, vertical line down for child. (In notes we use `M` / `F` and `→` / `↓`.)

Types

1. **Self-referential** — "Pointing to a photo: 'his father is my father's son' " → speaker is father of the photo-subject.
2. **Generation gap** — count levels between A and B.
3. **Cousin / in-law chain** — mother's brother's wife's son etc.
4. **Coded family relations** — P+Q means P is father of Q, etc. Decode with a key.
5. **Family tree puzzle** (grid-style) — several facts, build the full tree.
6. **"How is A related to B"** asked when tree is already built — just traverse.

Shortcut: build the tree once; answer every sub-question off it.

PART 5 — SEATING ARRANGEMENT

5.0 Opener

Types of arrangements: - Linear (n people in a row, some facing north, some south). - Circular (facing centre / facing outward / both). - Rectangular / square (4 on corners + middle). - Parallel rows (two rows facing each other). - Double-line (e.g., people + their gifts).

Attack plan (universal): take the **most-constrained** clue first; build a skeleton; plug in permissive clues last.

Types

1. **Pure linear, all facing same way**.
2. **Linear with mixed facing** — pay attention to left/right inversion.
3. **Circular, facing centre** — left/right of X is unambiguous.
4. **Circular, facing outward** — left/right flips.
5. **Mixed facing in circle** — track each person's orientation.
6. **Rectangular with 8 persons** — diagonals + sides rules.
7. **Two parallel rows** — "opposite", "immediate left of opposite" etc.
8. **Arrangement + attribute** (profession/color/age) — 2-variable puzzle.
9. **Arrangement + attribute + 2nd attribute** — 3-variable puzzle; backbone of banking mains.
10. **Month / day / floor puzzles**.
11. **Order + ranking** — (from top / from bottom, total count).

For multi-variable puzzles: make **one table per variable aligned to the seat skeleton**.

PART 6 — PUZZLES (FLOOR / BOX / SCHEDULING)

6.0 Opener

Same discipline as seating. The puzzle is a **constraint-satisfaction problem**; you are a human SAT solver.

Types

1. **Floor-based**: n persons, n floors; clues like "A lives immediately above B".
2. **Box stacking** — "box P is above Q but below R".
3. **Day-based scheduling** (Mon-Sun) — who works which day + extra constraint.
4. **Month-date combo** — 7 people, 7 months, 2 possible dates each.
5. **Colour / city / sport** — 3-dimensional attribute binding.
6. **Comparison puzzles** — A > B > C ranking across multiple metrics.
7. **Mixed floor + facing + attribute** — banking-mains boss level.
8. **Uncertain puzzles** — when the answer needs "cannot be determined".

Attack plan: Pick the clue that **fixes the most positions at once**. Use the table format: rows = positions, columns = attributes. Fill the unambiguous cells first.

PART 7 — MATHEMATICAL INEQUALITIES

7.0 Opener

Memorised: how inequality chains combine. - A > B ≥ C → A > C. (Strict wins.) - A ≥ B = C → A ≥ C. - A > B, B > C, C = D → A > D.

Types

1. **Direct inequality** — "if P > Q ≥ R < S, is P > S?" Note: R < S doesn't fix P vs S.
2. **Coded inequality** — @, #, &, $ stand for >, ≥, <, ≤, =. Decode, then apply rules.
3. **Statement + conclusions** (I & II, "only I", "both", "either") — logical conjunction.
4. **Double-decker conclusions** — inequality chains of length 5-6.

Shortcut: replace all coded symbols with actual inequality signs FIRST. Then scan for the two terms you need; confirm an unbroken chain.

PART 8 — SYLLOGISM

8.0 Opener

Memorised: 4 standard statements. - **A**: All S are P. - **E**: No S is P. - **I**: Some S are P. - **O**: Some S are not P.

And a fundamental fact: **"Some S are P" also means "Some P are S"** (conversion rule for I). "No S is P" converts to "No P is S" (conversion of E).

Types

1. **Two statements, two conclusions** — 'follows / does not follow'.
2. **Three statements, two conclusions**.
3. **Possibility conclusions** — "at least some" vs "all S are P is a possibility".
4. **Negative universal + particular** — mixing A/E/I/O.
5. **Chained syllogism** — A→B→C type conclusions.

Attack plan — the Venn-diagram method (universal shortcut): 1. For each statement, draw the unambiguous Venn region (Overlap for I, disjoint for E, subset for A, complement for O). 2. For each conclusion, check if ALL consistent Venn diagrams support it. 3. If even one valid Venn contradicts the conclusion, it "does not follow".

This is the one method that NEVER fails, unlike mnemonic trick-rules that break on uncommon cases.

PART 9 — INPUT-OUTPUT (Banking Mains)

9.0 Opener

A sequence of rearrangements (swap, shift, insertion, arithmetic op on numbers) is applied repeatedly; the candidate must find the rule and predict final / intermediate steps.

Types

1. **Word/number pure shifting** (e.g., words sorted alphabetically left-to-right).
2. **Arithmetic on numbers** (each number increased by index, squared, etc.).
3. **Odd-even splitting**.
4. **Combined word-number** (alphabetise AND square).
5. **New-age input-output** (complex rules — Bank PO mains speciality).

Shortcut: compare input and step 1 only first; detect the rule; verify on step 2 → step 3. Solve fast.

PART 10 — DATA SUFFICIENCY

10.0 Opener

Given a question + 2 (or 3) statements; determine which subset of statements suffices.

Answer format (standard SBI/IBPS): - (a) Statement I alone suffices. - (b) Statement II alone suffices. - (c) Each alone suffices. - (d) Both together needed. - (e) Neither alone nor together sufficient.

Types

1. **Quantitative DS** — solve with I only? with II only? with both?
2. **Reasoning DS** — similar but with ranking/seating/family tree statements.
3. **Syllogism / directional DS**.

Attack: deliberately IGNORE one statement while checking the other. Never merge too early.

PART 11 — NON-VERBAL REASONING (VISUAL)

> ***Diagram accuracy is mandatory here.*** *The types below are taught by method; shipped questions each have a to-scale SVG rendering verified before publication.*

11.1 Mirror Image (vertical line of reflection)

Rule: horizontally flip across a vertical mirror. Letters like A, H, I, M, O, T, U, V, W, X, Y read the same in mirror; B, C, D, E, K visibly flip; N, S, Z reverse along diagonals. Digits 0, 8 same; 1, 2, 3, 5, 6, 7, 9 change.

Shortcut: imagine writing the string, then read it in reverse order with each character's mirror-image shape.

11.2 Water Image (horizontal line of reflection)

Rule: vertical flip. Letters like B, C, D, E, H, I, K, O, X are visually symmetric under appropriate axes; most are not.

11.3 Paper Folding & Paper Cutting

Rule: a paper is folded k times along shown lines; then a cut of given shape is made; you must show how the unfolded paper looks.

Method: fold mentally in reverse. For each fold, reflect the cut about the fold-line. Do this k times.

11.4 Cubes & Dice

Core fact (opposite-face rule): on a standard die, opposite faces sum to 7. So 1↔6, 2↔5, 3↔4.

Types: - **Unfolded cube (net)** — identify which faces are opposite in the final assembled cube. - **Painted cube problem** — cube painted on outside, cut into n^3 small cubes; count small cubes with 3 / 2 / 1 / 0 painted faces. - Corners (always 8): 3 faces painted. - Edges (non-corner): $12·(n−2)$ cubes with 2 painted. - Face centres: $6·(n−2)^2$ cubes with 1 painted. - Interior: $(n−2)^3$ cubes with 0 painted. - **Dice rotation** — given two/three views, figure out which face lands where.

11.5 Embedded Figures

Find a given simple figure inside a complex one. Method: outline the target figure's defining lines; trace each possible position in the complex figure.

11.6 Figure Series

Series of figures following a rule (rotation by 45°, addition of one line per step, shading pattern). Method: isolate the **rule per element** (rotation, count, shade), then apply.

11.7 Analogy of Figures

A : B :: C : ? — find the transformation A→B and apply to C.

11.8 Classification of Figures

Identify the odd-one-out; usually by shape, count of angles, symmetry, or shading.

PART 12 — VERBAL REASONING (Statement-based)

Types

1. **Statement + Conclusion** — does the conclusion strictly follow?
2. **Statement + Assumption** — is the assumption necessary for the statement?
3. **Statement + Argument** — is the argument STRONG / WEAK?
4. **Statement + Course of action** — is the action appropriate?
5. **Cause and effect**.

These resemble CAT's verbal logical reasoning. Drill on past papers.

PART 13 — RANKING / ORDER

Types

1. **Single row ranking** — "A is 7th from left, 12th from right — total?"
2. **Before / after in queue**.
3. **Moved forward / backward** — "A moves 3 places up, now at 5th from top; earlier position?".
4. **Heights / weights ranking**.

Key formulas:

Scenario	Formula
Total = ? given rank from left (L) and right (R)	Total = L + R − 1
Rank from right given total (T) and rank from left (L)	R = T − L + 1
Position after moving k places up	New = Old − k

✎ WORKED EXAMPLE

Example. A is 7th from left and 12th from right in a row.

```
**Total = L + R - 1 = 7 + 12 - 1 = 18 people**
```

Example. In a class of 30, A moves 5 places up and is now 8th from top. Original rank?

```
**New rank = 8th → Old rank = 8 + 5 = 13th from top**
```

APPENDIX — Attack-plan matrix

Cue in stem	Part	Critical first move
"sitting in a row"	5	draw n-cell skeleton
"lives on floor"	6	draw n-floor skeleton (bottom = 1)
"pointing at photo"	4	identify speaker's relation first
"walks 5 m north, 3 m east"	3	draw on N-E cross
"all, some, no"	8	Venn diagrams
"mirror/water image"	11	check axis of reflection
"input-output steps"	9	diff step 1 vs input
"> , ≥ , < , ≤"	7	chain scan, no skipping
"Statement I / II, which"	10	test each alone first
"cube cut into 125 smaller"	11.4	n=5, apply corner/edge/face counts

APPENDIX B — Figure Asset Policy

No reasoning figure is shipped unless it is: 1. Authored as SVG with exact coordinates / angles. 2. Physically verified — a mirror is an actual horizontal reflection, not a lazy sketch. 3. Reviewed once against the answer — if the figure does not uniquely admit the "correct" option, it is rejected.

Crude ASCII-art diagrams are banned from this book and from shipped questions. If a figure asset is not ready, the corresponding question is held back until the asset is authored and verified.

PART 14 — WORKED-EXAMPLES DEEP DRILL

*One fully-worked specimen per high-yield type. Memorise the **method**, not the specific example.*

14.1 Number series — second-difference method

✎ WORKED EXAMPLE

Q. Find the missing term: 4, 9, 17, 28, ?, 57.

```
**Step 1 — First differences (Δ¹):** 5, 8, 11, ?, ? **Step 2 — Second differences (Δ²):** 3, 3, 3 → constant! (Arithmetic progression) **Step 3 — Next Δ¹:** 11 + 3 = **14** **Step 4 — Missing term:** 28 + 14 = **42** **Step 5 — Verify last:** 42 + 15 = 57 ✓ (Δ¹ = 14, next Δ² = 3 → Δ¹ = 15)
```

Answer: 42 **Universal rule:** Compute Δ^1 first. If constant → AP. If not, compute Δ^2. If Δ^2 is constant → quadratic. If Δ^2 is also not constant → try ratio, squares, or cubes.

× COMMON TRAP

When the series seems wrong: If your derivation gives an answer that isn't among the options, try the next simplest rule. In exams, the series always has an elegant rule — you haven't found it yet.

14.2 Letter series — position method

✎ WORKED EXAMPLE

Q. B, D, G, K, ?

```
**Step 1 — Convert to positions (A=1, B=2, …):** B=2, D=4, G=7, K=11, ? **Step 2 — First differences:** 2, 3, 4, → next = **5** **Step 3 — Next position:** 11 + 5 = 16 = **P**
```

Answer: P

14.3 Coding-decoding — find the rule first

✎ WORKED EXAMPLE

Q. TIGER is coded as UJHFS. How is LION coded?

> **Step 1 — Find the rule:** | Original | T | I | G | E | R | |---|---|---|---|---|---| | Position | 20 | 9 | 7 | 5 | 18 | | Coded | U | J | H | F | S | | Coded pos | 21 | 10 | 8 | 6 | 19 | | Shift | +1 | +1 | +1 | +1 | +1 | **Rule: Each letter shifted +1 position forward** **Step 2 — Apply to LION:** | L(12) | I(9) | O(15) | N(14) | |---|---|---|---| | M(13) | J(10) | P(16) | O(15) |

Answer: MJPO

14.4 Direction sense — net displacement

✎ WORKED EXAMPLE

Q. A walks 4 km North, 3 km East, 4 km South, 6 km West. Distance from start?

> **Step 1 — Net North-South:** 4 km N - 4 km S = **0** (cancelled) **Step 2 — Net East-West:** 3 km E - 6 km W = **3 km West** (net) **Step 3 — Distance from start:** $\sqrt{0^2+3^2}$ = **3 km due West**

Pythagorean variation: 6 km North then 8 km East → distance = $\sqrt{6^2+8^2} = \sqrt{36+64} = \sqrt{100}$ = **10 km**

14.5 Blood relations — trace step by step

✎ WORKED EXAMPLE

Q. A woman says about a man: "He is the only son of my mother's husband's mother." How is the man related to her?

```
**Step 1:** "Mother's husband" = her **father** **Step 2:** "Father's mother" = her **grandmother** **Step 3:** "Only son of grandmother" = her **father** (no uncles; only son) **Answer:** Man is her **father**
```

✕ COMMON TRAP

Trap: Students often stop at "grandmother's son" and call him "uncle". Re-read "only son" — this explicitly eliminates brothers, making him the father.

14.6 Seating arrangement (circular facing centre)

✎ WORKED EXAMPLE

Q. A, B, C, D, E, F sit in a circle facing the centre. A is 3rd to the left of D. B is 2nd to the right of E. C sits between A and F. F is not adjacent to D. Find the order. **Convention:** Facing centre → your LEFT is clockwise in the drawn diagram.

```
**Step 1 — Most constrained clue first:** "A is 3rd to left of D" Fix D at position 1. 3 places left (clockwise) = position 4. **A = 4**. **Step 2 — Next clue:** "B is 2nd to the right of E" Remaining positions: 2, 3, 5, 6 for B, C, E, F. If E = 2 → B = 4 (taken by A). Invalid. If E = 3 → B = 5. ✓ If E = 5 → B = 1 (taken by D). Invalid. If E = 6 → B = 2. ✓ **Step 3 — Check "F not adjacent to D (pos 1)":** F cannot be at 2 or 6. If E=3, B=5: remaining {2, 6} for C, F. F ≠ 2 and F ≠ 6 → IMPOSSIBLE. If E=6, B=2: remaining {3, 5} for C, F. F can be 3 or 5. ✓ **Step 4 — "C sits between A(4) and F":** With E=6, B=2: C and F at 3 and 5. Between A(4) and the open seats: pos 3 is between A(4) and B(2); pos 5 is between A(4) and E(6). C between A and F: if F=5, C must be at 3 (between A and... no, F is at 5 which IS next to A). Try F=3, C=5. C(5) is between A(4) and E(6) ✓, and F(3) is not adjacent to D(1) ✓. **Final arrangement (CW):** D(1), B(2), F(3), A(4), C(5), E(6)
```

✓ KEY FACT

Method takeaway — circular seating: 1. Draw 6 numbered seats in a circle (clockwise). 2. Fix the most constrained person first. 3. Translate "left"/"right" using the convention: facing centre → left = clockwise. 4. Test each possibility systematically — eliminate contradictions.

14.7 Floor puzzle — systematic elimination

✎ WORKED EXAMPLE

Q. P, Q, R, S, T live on floors 1–5 (1 = ground, 5 = top). T = topmost. S = even floor. R lives between S and T. P lives above Q.

```
**Step 1:** T = floor 5 (given directly) **Step 2:** S = even → S = 2 or 4 **Step 3:** "R between S and T" means S < R < 5 (strictly between) - If S = 4: R must be between 4 and 5 — no integer possible → **S ≠ 4** - So S = **2**, R = 3 or 4 **Step 4:** Remaining people P, Q get floors {1, 4} or {1, 3}. "P above Q" → P > Q | R | Remaining | P above Q | |---|---|---| | R = 3 | P, Q ∈ {1, 4} → P = 4, Q = 1 | ✓ | | R = 4 | P, Q ∈ {1, 3} → P = 3, Q = 1 | ✓ | Both satisfy constraints. If the exam question asks "Who is on floor 4?" — only R=3 arrangement gives a unique answer (P=4).
```

14.8 Coded inequality — always decode first

✎ WORKED EXAMPLE

Q. Symbols: @ = >, # = ≥, & = <, $ = ≤, * = =. Statement: **P @ Q # R & S * T**
Conclusions: (I) P > T (II) S < P

```
**Step 1 — Decode the chain:** P > Q ≥ R < S = T **Step 2 — Check conclusion I (P > T):** Trace: P > Q ≥ R … then R **< S** — direction reverses here. The chain breaks at "<". We cannot compare P and S (or T). **I does not follow.** **Step 3 — Check conclusion II (S < P):** S = T (from * =). But S > R, and P > Q ≥ R — does not tell us S vs P. Chain broken (same break point). **II does not follow.**
```

Answer: Neither conclusion follows.

✔ KEY FACT

The golden rule: Any chain that contains BOTH > (or ≥) AND < (or ≤) has a direction reversal — no definite conclusion about the endpoints.

14.9 Syllogism — the Venn method

✎ WORKED EXAMPLE

Q. Statements: (A) All cats are dogs. (B) Some dogs are bats. Conclusions: (I) Some cats are bats. (II) Some bats are dogs.

```
**Step 1 — Draw Venn:** - "All cats are dogs" → cats circle is **inside** the
dogs circle - "Some dogs are bats" → dogs and bats circles **overlap**
(somewhere) **Step 2 — Check Conclusion I (Some cats are bats):** The overlap
of dogs & bats COULD be entirely outside the cats region. Not forced by the
statements → **I does not follow.** **Step 3 — Check Conclusion II (Some bats
are dogs):** "Some dogs are bats" directly converts to "Some bats are dogs"
(I-type conversion). This is always valid → **II follows.**
```

Answer: Only II follows.

× COMMON TRAP

Never assume that because "All cats are dogs" and "Some dogs are bats", cats must overlap with bats. The bat-dog overlap might be in the dog-only area (outside cats). Always draw the Venn before concluding.

14.10 Input-output — detect the rule from Step 1

✎ WORKED EXAMPLE

Q. Input: 6 19 12 3 15. Step 1: 7 21 15 6 19. Find Step 2.

```
**Step 1 — Compare input vs Step 1 position by position:** | Position | 1 | 2
| 3 | 4 | 5 | |---|---|---|---|---|---| | Input | 6 | 19 | 12 | 3 | 15 | |
Step 1 | 7 | 21 | 15 | 6 | 19 | | Change | +1 | +2 | +3 | +3 | +4 |
**Observation:** Each number at position i gets +i added to it. **Step 2 —
Apply the same rule to Step 1:** | Position | 1 | 2 | 3 | 4 | 5 | |---|---|---
|---|---|---| | Step 1 | 7 | 21 | 15 | 6 | 19 | | Add | +1 | +2 | +3 | +4 | +5
| | Step 2 | **8** | **23** | **18** | **10** | **24** |
```

14.11 Data sufficiency (Type 1)

Q. Find the value of x. (I) x + 5 = 12. (II) 2x = 14.

Method. - I alone: $x = 7$. **Sufficient.** - II alone: $x = 7$. **Sufficient.**

Answer: **Each statement alone is sufficient (option c).**

14.12 Painted cube (Type 11.4)

Q. A cube of side 5 cm is painted on all faces, then cut into 1 cm^3 smaller cubes. How many small cubes have exactly 2 faces painted?

Method. Edge cubes (excluding corners): $12 \cdot (n - 2)$ where $n = 5 \rightarrow 12 \cdot 3 =$ **36**.

> *Memorise: corners 8 (3-painted), edges $12(n-2)$ (2-painted), face centres $6(n-2)^2$ (1-painted), interior $(n-2)^3$ (0-painted).*

PART 15 — BANKS-FOCUS DEEP DRILL

15.1 Banking reasoning weight (per section)

Topic	Prelims	Mains
Puzzles (linear / circular / floor / box / scheduling)	10-15	12-18
Seating arrangement	5-10	8-12
Syllogism (incl. possibility)	3-5	3-5
Coded inequality	2-5	3-5
Blood relations (incl. coded)	2-3	3-5
Direction (incl. coded)	2-3	2-3
Input-output	0 (rare)	4-5
Data sufficiency	2-3	3-5
Logical (statement + conclusion / argument)	—	3-5
Misc (alphanumeric, ranking)	2-5	—

*So a Banks aspirant must over-prepare **puzzles + seating + syllogism + coded inequality + DS + input-output**. These give >70% of the section.*

15.2 The 4 banking-puzzle archetypes (drill these to mastery)

1. **Linear single-row (n persons, mixed facing)** — clue types: "second to left of X", "between Y and Z facing same direction".
2. **Linear two-parallel-rows** — "opposite", "diagonally opposite", "to the left of person opposite to X".
3. **Circular (facing centre or outward, with one mixed)** — left/right reverses on outward facers.
4. **Floor / box / scheduling combo** — month + day + person, or floor + flat-number + colour.

*For each type, the **first move** is identical: write the skeleton with all positions blank. Pick the most-constrained clue (one that fixes ≥2 positions). Fill iteratively. If a contradiction arises, backtrack to the most recent unforced choice.*

15.3 Advanced syllogism — possibility conclusions

Banks Mains often asks "All A being C is a possibility" — the candidate must check if at least ONE valid Venn diagram allows the universal A⊆C while satisfying the given statements.

Q. Statements: Some pens are pencils. All pencils are erasers. Conclusion: All pens being erasers is a possibility.

Method. Draw — pens overlap pencils, pencils inside erasers. The portion of pens NOT overlapping pencils may or may not be inside erasers — the statements don't restrict this. So we CAN draw a diagram where all pens are inside erasers. **Possibility holds.** Answer: **conclusion follows.**

> *Possibility-rule: as long as no statement contradicts the proposed Venn, the possibility holds.*

15.4 Machine input-output (banking mains specialty)

A machine produces a sequence of steps that rearrange/transform an input list. Each step applies a specific rule. The candidate is given input + first 1-2 steps and must predict step 4 / step 5 / answer questions on intermediate steps.

Drill rule. Compare input ↔ step-1 differences first: - Sorted alphabetically? Sorted numerically? Reversed? - Each number squared? Cubed? Index-added? - Words and numbers separated? - Two halves (left/right) treated differently?

Common rules in 2024-26 papers: - **Index-shift** — ith element shifts by i positions. - **Sum-of-digits replacement** — each number → sum of its digits. - **Alphabet position swap** — words turn into the alphabet position of their first letter. - **Squares of even, cubes of odd**.

15.5 Coded inequality (advanced — 2-statement conjunction)

Banks Mains gives 2 statements with codes, then 4 conclusions. Methods: 1. Decode each statement separately. 2. For each conclusion, check both statements for a chain proof. 3. Use "either–or" — sometimes conclusions I and II share a common element where exactly one must be true (e.g., if I says "P > Q" and II says "P = Q", and the chain doesn't decide, then "either I or II" is the answer).

15.6 Data sufficiency — 3-statement variant

Some banks (RBI, IBPS PO Mains) ask 3-statement DS: - (a) Only I and II. - (b) Only II and III. - (c) All three. - (d) Any two. - (e) All needed.

Discipline: test pairs (I+II, I+III, II+III) before testing all three. Often two suffice.

15.7 30-day Banks reasoning plan

Days	Topic	Hours/day
1-3	Inequality (basic + coded)	1
4-7	Syllogism (Venn + possibility)	1.5
8-12	Linear seating (single + two-row)	2
13-17	Circular seating + mixed facing	2
18-22	Floor / box / scheduling puzzles	2
23-25	Blood relations + direction	1
26-28	Input-output + DS	1.5
29-30	Mock + weak-area review	3

PART 16 — SSC + RRB FOCUS DEEP DRILL

16.1 SSC + RRB reasoning weight (per section)

Topic	SSC CGL T1	SSC CHSL T1	RRB NTPC CBT 1	RRB NTPC CBT 2
Series (number, letter, alphanumeric)	3-4	3-4	3-5	3-5
Analogy (word, letter, number, figure)	3-5	3-5	4-6	4-6
Classification (odd-one-out)	2-3	2-3	3-4	3-4
Coding-decoding	2-3	2-3	3-4	3-4
Blood relations	2-3	1-2	2-3	2-3
Direction	1-2	1-2	1-2	1-2
Mathematical operations / DI	2-3	2-3	2-3	2-3
Mirror/water image (non-verbal)	1-2	1-2	1-2	1-2
Cube/dice	1-2	1-2	0-1	0-1
Embedded figures	1	1	0-1	0-1
Paper folding / cutting	1-2	1-2	0	0
Syllogism (basic)	1-2	1-2	1-2	1-2
Word formation / alphabet	1-2	1-2	1-2	1-2
Misc (Venn, missing letters/numbers)	1-2	1-2	1	1

*So an SSC/RRB aspirant must over-prepare **series + analogy + classification + coding + non-verbal**. Banks-style heavy puzzles are LESS frequent.*

16.2 Non-verbal mastery — the 4 must-knows

1. **Mirror image** — vertical mirror, horizontal flip of each glyph; sequence reversed too.
2. **Water image** — horizontal mirror, vertical flip of each glyph (digits 0, 1, 8 unchanged; B, D, E, H, I, K, O, X have horizontal-symmetric letters).
3. **Paper folding** — fold in reverse; each cut reflects across the fold line per fold.

4. **Cube assembly from net** — opposite faces never share an edge in the unfolded net. Use the "opposite-face rule" (1↔6, 2↔5, 3↔4).

16.3 Worked: Mirror image

Q. What is the mirror image (vertical axis) of the word "HORSE"?

Method. Reverse the order of letters and mirror each glyph individually: - Reversed: ESROH. - Mirror each: E → Ǝ; S → 2 (mirror); R → Я; O → O; H → H. - Result: **ƎƧЯOH** (reading left-to-right in mirror).

> *Trap: don't just reverse — you must mirror each character too. H, O, X, M, etc. are self-symmetric; B, C, D, E, K are not.*

16.4 Worked: Cube net

Q. A cube is unfolded as a "cross" net with faces labelled 1-6. Face 1 is centre. Faces 2, 3, 4, 5 are around face 1 (cross arms). Face 6 is at the far end of one arm. Which face is opposite face 6?

Method. Adjacent faces in the net never end up opposite. Face 6 is at the tip of an arm, so its **immediate** neighbour in the net (the face attached to it) becomes opposite face 1's opposite — actually, the standard rule: in a cross net, **the centre face's opposite is the far-tip face on the longest arm**. Tip face = 6. Centre = 1. So 1 ↔ 6.

> *Memorise the cross-net rule: arms-of-equal-length give arm-ends opposite each other.*

16.5 Analogy (word + number) — common patterns

Pattern	Example
Squares	4 : 16 :: 7 : 49
Cubes	3 : 27 :: 5 : 125
n + 1, n − 1	5 : 6 :: 9 : 10
Multiplication by k	4 : 12 :: 7 : 21 (×3)
Letter pair shifted by k	A : C :: M : O (+2)
Reverse letter	A : Z :: B : Y
Synonym / antonym	Happy : Joyful :: Sad : Sorrowful
Tool : worker	Knife : Chef :: Brush : Painter

Pattern	Example
Animal : young one	Dog : Puppy :: Horse : Foal
Country : currency	India : Rupee :: Japan : Yen
Country : capital	India : Delhi :: France : Paris
Author : book	Tagore : Gitanjali :: Kalidasa : Meghaduta
Cause : effect	Storm : damage :: Drought : famine

16.6 SSC speed rules (60 min, 25 questions)

- **Series + analogy + classification** — 12-15 questions. Target 25-30 sec each.
- **Coding + blood-relations + direction** — 6-8 questions. Target 40-60 sec each.
- **Non-verbal + cubes + mirror** — 3-5 questions. Target 30-45 sec each.
- Total budget: ~15 minutes for the 25-question reasoning section. Faster Q-types front-load; non-verbal in the middle (visual-fresh); save heaviest puzzle for last.

16.7 30-day SSC reasoning plan

Days	Topic	Hours/day
1-3	Number + letter + alphanumeric series	1
4-6	Analogy (word + letter + number)	1
7-9	Classification + odd-one-out	1
10-12	Coding-decoding (all variants)	1.5
13-15	Blood relations + direction	1.5
16-18	Math operations + Venn	1
19-21	Mirror, water, embedded figures	1.5
22-24	Paper folding + cube/dice	1.5
25-27	Syllogism (basic) + ranking	1
28-30	Mock + speed-drill weak topics	3

PART 17 — UNIVERSAL ATTACK CHECKLIST (1-page revision card)

Before the section starts (in your head)

1. Compass rose drawn on rough sheet. Alphabet row + index row written down.
2. Family-tree symbols ready (M / F / → / ↓).
3. Cube opposite-face rule (1↔6, 2↔5, 3↔4).
4. Squares 1-30, cubes 1-15 cached.
5. Inequality conversion rules: > beats ≥ beats =.
6. Venn-method ready for syllogism.

During each question

Cue word	Topic	Action
series, complete the	Series	Δ row
code, written as	Coding	letter-shift table
walks, faces, turns	Direction	N-E cross, vector-add
relation, related	Blood relations	family tree
sit in a row	Linear seating	n-cell skeleton
sit around table	Circular seating	n-position circle, lock most-constrained
live on n floors	Floor puzzle	n-floor skeleton, 1=ground
@, #, &, $, *	Coded inequality	decode → chain scan
All / Some / No	Syllogism	Venn
input, after step	Input-output	input ↔ step 1 first
Statement I / II	Data sufficiency	each alone first
mirror / water / fold / dice	Non-verbal	apply method per § 11

APPENDIX — Attack-plan matrix

Cue in stem	Part	Critical first move
"sitting in a row"	5	draw n-cell skeleton
"lives on floor"	6	draw n-floor skeleton (bottom = 1)
"pointing at photo"	4	identify speaker's relation first
"walks 5 m north, 3 m east"	3	draw on N-E cross
"all, some, no"	8	Venn diagrams
"mirror/water image"	11	check axis of reflection
"input-output steps"	9	diff step 1 vs input
"> , ≥ , < , ≤"	7	chain scan, no skipping
"Statement I / II, which"	10	test each alone first
"cube cut into 125 smaller"	11.4	n=5, apply corner/edge/face counts

APPENDIX B — Figure Asset Policy

No reasoning figure is shipped unless it is: 1. Authored as SVG with exact coordinates / angles. 2. Physically verified — a mirror is an actual horizontal reflection, not a lazy sketch. 3. Reviewed once against the answer — if the figure does not uniquely admit the "correct" option, it is rejected.

Crude ASCII-art diagrams are banned from this book and from shipped questions. If a figure asset is not ready, the corresponding question is held back until the asset is authored and verified.

Use this book alongside the Arithmetic and English books for complete skill coverage.

Practice Set (50 exam-style worked examples)

> *Past-paper analysis for each item: stem → smart method → answer → trap.*

Section 1 — NUMBER + LETTER SERIES (5 worked + variations)

> ***Notation expansion (used throughout): Δ (delta)*** *= first difference between consecutive terms.* ***Δ^2 (delta-squared)*** *= second difference (difference of Δ values). Constant Δ → arithmetic progression; constant Δ^2 → quadratic series.* ***Alphabet position:*** *A=1, B=2, ..., Z=26. Memorise: M=13, N=14, O=15, P=16.*

Q1. Find the next term: 3, 8, 15, 24, ?

What kind of problem is this? Number series — first compute the differences between consecutive terms.

Solving it. Step-by-step. - Differences: 8−3=5; 15−8=7; 24−15=9 → Δ = 5, 7, 9, ? - Δ-of-Δ (Δ^2) = 2, 2 → constant. So next Δ = 9 + 2 = **11**. - Next term = 24 + 11 = **35**.

Worth knowing: When Δ values increase in a constant way (here +2 each time), the underlying series is **n^2 + something**. Check: terms = 3, 8, 15, 24 → these are $n^2 + (n-1)$ for n = 2, 3, 4, 5. Wait: 4+(-1)=3 ✓; 9+(-1)=8 ✓; verify $n^2 - 1$ for n=2,3,4,5 → 3, 8, 15, 24 ✓. So pattern is $n^2 - 1$ for n=2,3,4,5,6 → 6th = 36 − 1 = **35** ✓.

Watch out: "33" — students extrapolate Δ as 5, 7, 9, 11 ✓ but compute 24 + 9 = 33 by mistake (using current Δ instead of next Δ).

A similar question — Square minus 1. 0, 3, 8, 15, 24, 35, ? → next term: n=7 → 49 − 1 = **48**.

Another twist — Cubic series. 1, 8, 27, 64, ? → cubes; next n=5 → **125**.

Q2. Find the next term: 2, 6, 12, 20, 30, ?

What kind of problem is this? Number series — recognise the n(n+1) pattern (product of two consecutive integers).

Solving it. Step-by-step. - 2 = 1×2; 6 = 2×3; 12 = 3×4; 20 = 4×5; 30 = 5×6 - Pattern: nth term = n(n+1) - 6th term = 6×7 = **42**

Quick way (if pattern not spotted). Differences: 4, 6, 8, 10, ?. $\Delta^2 = 2$ (constant). Next $\Delta = 12$. Next term = 30 + 12 = **42** ✓.

Watch out: "40" — students assume Δ keeps growing by 4 (4, 6, 8, 10, 12 looks right) but they apply 30 + 10 = 40 instead of 30 + 12.

A similar question — Triangular numbers. 1, 3, 6, 10, 15, 21, ? → these are n(n+1)/2. Next = 7×8/2 = **28**.

Another twist — n^2+n+1. 3, 7, 13, 21, 31, ? → Δ = 4, 6, 8, 10. Next Δ = 12. Next = 31 + 12 = **43**.

Q3. Find the next term: A, C, F, J, O, ?

What kind of problem is this? Letter series — convert to alphabet positions, then find numerical pattern.

Solving it. Step-by-step. - A=1, C=3, F=6, J=10, O=15. - Differences: 2, 3, 4, 5 → next = 6. - Next position = 15 + 6 = **21 → letter U**.

Worth knowing: Memorise key alphabet positions cold: A=1, E=5, J=10, O=15, T=20, Z=26.

Watch out: Students count alphabet positions wrong (e.g., O=14 instead of 15).

A similar question — Backward alphabet. Z, X, U, Q, ? → positions 26, 24, 21, 17, ? → diffs 2, 3, 4, ? → next diff 5; next position 17−5=12 → **L**.

Another twist — Skip with offset. B, D, G, K, P, ? → positions 2, 4, 7, 11, 16, ? → diffs 2, 3, 4, 5, 6 → next position 16+6=22 → **V**.

Q4. Find the next term: 5, 11, 23, 47, 95, ?

What kind of problem is this? Multiplicative-recursive series. Each term related to the previous by a fixed rule.

Solving it. Step-by-step. - Test Δ: 6, 12, 24, 48, ? → doubles each time. So next Δ = 96. Next term = 95 + 96 = **191**. - Verify alternate rule: each term = 2 × prev + 1? - 2(5)+1 = 11 ✓; 2(11)+1 = 23 ✓; 2(23)+1 = 47 ✓; 2(47)+1 = 95 ✓; 2(95)+1 = **191** ✓.

Watch out: "190" — students compute 95 × 2 = 190 and forget the +1.

A similar question. 1, 3, 7, 15, 31, ? → each = 2×prev + 1. Next = 2(31)+1 = **63**.

Another twist — 2x – 1 pattern. 3, 5, 9, 17, 33, ? → each = 2×prev – 1. Next = 2(33)−1 = **65**.

Q5. Find the next term: 1, 4, 9, 16, 25, ?

What kind of problem is this? Pure squares.

Solving it. $1=1^2$, $4=2^2$, $9=3^2$, $16=4^2$, $25=5^2$. Next = 6^2 = **36**.

A similar question — Square + 1. 2, 5, 10, 17, 26, ? → n^2+1. Next = 6^2+1 = **37**.

Another twist — Reverse squares. 49, 36, 25, 16, 9, ? → 7^2, 6^2, 5^2, 4^2, 3^2, ? → next = 2^2 = **4**.

Section 1A — ADVANCED SERIES (exam-level variations)

Q5.5 (Advanced) — Mixed-operations series. 8, 17, 35, 71, 143, ?

Solving it. Test relation between consecutive terms. - 8 → 17: 8 × 2 + 1 = 17 ✓ - 17 → 35: 17 × 2 + 1 = 35 ✓ - 35 → 71: 35 × 2 + 1 = 71 ✓ - 71 → 143: 71 × 2 + 1 = 143 ✓ - Next: 143 × 2 + 1 = **287**

Worth knowing: Other "× a + b" patterns to test: ×2−1, ×3+2, ×3−1, etc.

Q5.6 (Advanced) — Alternating series (two interwoven patterns). 2, 5, 4, 11, 6, 17, 8, 23, ?

Solving it. - Odd positions (1st, 3rd, 5th, 7th): 2, 4, 6, 8 → AP with d = 2. Next: 10. - Even positions (2nd, 4th, 6th, 8th): 5, 11, 17, 23 → AP with d = 6. Next: 29. - 9th position is odd → next term = **10**.

Watch out: Treating it as a single sequence; differences won't be constant.

Q5.7 (Advanced) — Number + symbol series. A1, C4, F9, J16, O25, ?

Solving it. Decode letter and number separately. - Letters: A(1), C(3), F(6), J(10), O(15) — diffs 2, 3, 4, 5 → next 6 → 15+6 = 21 → **U**. - Numbers: 1, 4, 9, 16, 25 = squares → next = 36. - Answer: **U36**.

Section 2 — CODING-DECODING (5 worked + variations)

> ***Notation expansion: Coding*** *= applying a rule (shift, position, sum, reverse) to map letters → digits OR letters → letters. Always* ***write the alphabet row + position row first*** *on rough paper. This eliminates 80 % of errors. Common shift rules: +1, +2, +k; reverse (Z=1, A=26); position-dependent; block-shift.*

Q6. If ROSE is coded as 6821 and CHAIR as 73456, what is the code for SEARCH?

What kind of problem is this? Letter-to-digit substitution mapping. Build the dictionary from the given codes, then apply.

Solving it. Step-by-step.

Step 1 — Extract mappings from given. - ROSE → 6821: R=6, O=8, S=2, E=1. - CHAIR → 73456: C=7, H=3, A=4, I=5, R=6 (R confirms = 6 ✓).

Step 2 — Look up each letter of SEARCH. - S=2, E=1, A=4, R=6, C=7, H=3. - SEARCH = 2-1-4-6-7-3 = **214673**.

Worth knowing: When a letter appears in both codes, it MUST map to the same digit — use this for verification.

Watch out: Students don't write the mapping table first; instead try to "remember" mappings from prose, leading to errors.

A similar question. If ROAD = 1234 and TANK = 5267, find the code for KNAT. Map: R=1, O=2, A=3, D=4; T=5, N=6, K=7 (A=3, K=7, N=6, T=5 ⇒ confirms). KNAT = 7-6-3-5 = **7635**.

Another twist — Reversal. If ROSE → ESOR (just reverse the order of letters). Then SEARCH → HCRAES.

Q7. If TIGER → UJHFS (each letter shifted by +1), then DONKEY = ?

What kind of problem is this? Uniform letter shift. Apply +k to each letter.

Solving it. Step-by-step. Apply +1 to each letter of DONKEY. - D → **E** - O → **P** - N → **O** - K → **L** - E → **F** - Y → **Z** - DONKEY = **EPOLFZ**.

Worth knowing: Wrap-around: Z + 1 = A (alphabet is cyclic). So if we had ZEBRA + 1 = AFCSB.

Watch out: Students shift in the wrong direction (−1 instead of +1) → CNMJDX (wrong).

A similar question — Shift by +2. TIGER + 2 = VKIGT.

Another twist — Shift backward. If LION → KHNN (each −1), then BEAR → ADZQ.

Q8. If CAT = 24 and DOG = 26, find the code for LION.

What kind of problem is this? Sum of alphabet positions.

Solving it. Step-by-step. - C=3, A=1, T=20 → sum = 24 ✓. - D=4, O=15, G=7 → sum = 26 ✓. - L=12, I=9, O=15, N=14 → sum = **50**.

Watch out: "59" — letter-position errors (students often miscount T as 19 or O as 16).

A similar question — Reverse-position sum. Use Z=1, Y=2, ..., A=26. CAT (reverse): C=24, A=26, T=7 → sum = 57.

Another twist — Product instead of sum. TEN = 20×5×14 = 1400.

Q9. If RED is coded as 729-125-64, find the code for CODE.

What kind of problem is this? Code uses **cubes** of values. Identify the base mapping first.

Solving it. Step-by-step. - 729 = 9^3, 125 = 5^3, 64 = 4^3. - For RED → R = ?, E = 5 ✓ (E=5 in alphabet, cube = 125 ✓), D = 4 ✓ (D=4, cube = 64 ✓). - For R → 9, but R's alphabet position = 18, not 9. So mapping is **reverse position**: R → 27 − 18 = 9. Verify D: 27 − 4 = 23, but cube (D=4) gave 64 not 23^3. Hmm — D's normal position 4 gives cube 64 ✓, so D uses normal position, not reverse. - The setup is INCONSISTENT — only R's "9" is reverse-position, while D's "4" and E's "5" are normal-position. Question is malformed.

Lesson. Always verify the rule on ALL given codes before applying. If the rule doesn't fit consistently, mark "data inconsistent" and move on.

Q10. In a code: PEN = 35, INK = 38, find BOOK.

What kind of problem is this? Possibly sum of positions; verify the rule.

Solving it. Step-by-step. - PEN: P=16, E=5, N=14 → sum = 35 ✓ (matches). - INK: I=9, N=14, K=11 → sum = 34 (NOT 38). - Sum rule fails for INK. Try positions + 4? 34 + 4 = 38 ✓ — but PEN gave 35 (already correct without +4). Inconsistent rule.

Lesson. Same as Q9 — always test the rule on BOTH given codes before applying. If inconsistent, the question is flawed. Mark "rule cannot be determined".

Section 2A — ADVANCED CODING (exam-level variations)

Q10.5 (Advanced) — Position-dependent coding. If "EARTH" is coded as "GFTVH", find the code for "MOON".

Solving it. Find shift per position. - E(5) → G(7): +2; A(1) → F(6): +5; R(18) → T(20): +2; T(20) → V(22): +2; H(8) → H(8): 0? - Re-check: maybe shifts are +2, +5, +2, +2, 0 — irregular. Try other rule. - Position-based: shift = position × k. Position 1 shift +2, position 2 shift +5 (not multiple). - Try: each letter shifted by +2, +5, +2, +2, 0 — irregular; question may use mixed rule (test on both given codes if 2 given).

Lesson: With only ONE coded example, multiple rules can fit. Banks Mains usually gives 2-3 examples.

Q10.6 (Advanced) — Block coding. "GOOD MORNING" coded as "TBYY EQDARZW". Find code for "EVENING".

Solving it. Decode letter-by-letter. - G→T: +13; O→B: +13 (O=15, +13 = 28 mod 26 = 2 = B); O→Y: +10? Inconsistent. - Try ROT-13 (shift +13, the classic cipher): - G(7)+13 = 20 = T ✓ - O(15)+13 = 28-26 = 2 = B ✓ - O(15)+13 = B (NOT Y) — discrepancy. - Maybe block-shifted: first word shifted +13, second word shifted differently. - M(13)+13=Z; O(15)+13=B; R(18)+13=E... if MORNING → ZBEAVAT. Doesn't match EQDARZW either. - Lesson: complex ciphers; always derive shift on first 2 letters and verify.

Q10.7 (Advanced) — Chinese-coding (symbol mapping table). Mapping: cat = ⊕; dog = ⊗; fish = ▲; black = ◇; brown = ◆; quick = ★. Phrase: "quick brown fish" coded as "★ ◆ ▲". Find code for: "black cat".

Solving it. Look up: black = ◇; cat = ⊕. **Answer:** ◇ ⊕.

Section 3 — BLOOD RELATIONS (5 worked + variations)

> ***Notation expansion:*** *Drawing a family tree on rough paper saves 80 % of errors. Use **M** for male, **F** for female; **–** for spouse; **|** for parent-child link. Generation count: parent → child = 1 step down; sibling = same generation; grandparent → grandchild = 2 steps down.*

Q11. Pointing to a man, Sita said, "He is the only son of my mother's mother." How is the man related to Sita?

What kind of problem is this? Self-referential relation chain. Trace each phrase carefully.

Solving it. Step-by-step. - "My mother's mother" = Sita's grandmother (maternal). - "Only son of [grandmother]" = Sita's grandmother's son. Since the speaker says "ONLY son", grandmother has just one son. - The grandmother's son (and Sita's mother's brother) = Sita's **maternal uncle**.

Worth knowing: Phrase decoder. - "Mother's mother" = maternal grandmother - "Father's father" = paternal grandfather - "Mother's brother" = maternal uncle (mama in Hindi) - "Father's brother" = paternal uncle (chacha) - "Mother's sister" = maternal aunt (mausi) - "Father's sister" = paternal aunt (bua/phupi)

Watch out: "Father" — students misread "mother's mother" as just "mother" and conclude the grandmother's only son = Sita's father (skipping a generation).

A similar question. Pointing to a woman, Ravi said, "She is the daughter of my father's only daughter." How is the woman related to Ravi? — "Father's only daughter" = Ravi's sister (since only daughter, no other sisters; Ravi himself is the brother). The woman = Ravi's sister's daughter = **Ravi's niece**.

Another twist. "He is the brother of my mother's only son." → Mother's only son = speaker himself. Brother of speaker = also a son of speaker's mother. The man is speaker's brother (only if mother has another son besides the speaker — contradicts "only"). So: man and speaker share mother → man IS the speaker, OR he must be a step-brother. (Often a trick — answer = "himself".)

Q12. A is B's brother. C is A's mother. D is C's father. E is B's son. How is E related to D?

What kind of problem is this? Multi-generational tree. Build it step by step.

Solving it. Step-by-step. - A and B are brothers (same parents). - C is A's (and therefore B's) mother. - D is C's father → D is A's and B's grandfather. - E is B's son → E is C's grandson → E is D's **great-grandson**.

Worth knowing: Generation count. - A → C: 1 generation up. - C → D: 1 generation up. So D is 2 generations above A. - A → B: same generation; B → E: 1 generation down. E is 1 below A. - So E is 3 generations below D → great-grandchild.

Watch out: "Grandson" — off by one generation. Always count generations carefully.

A similar question. P is Q's son. Q is R's daughter. R is S's mother. How is P related to S? P → Q → R → S means P is 1 below Q, Q is 1 below R, R is 1 below S. So P is 3 generations below S → P is S's **great-grandchild**.

Q13. P + Q means P is the father of Q. P − Q means P is the mother of Q. P × Q means P is the brother of Q. Decode: M + N − R × S. How is M related to S?

What kind of problem is this? Symbolic family code. Decode each operator, then build the chain.

Solving it. Step-by-step. - M + N: M is the father of N. - N − R: N is the mother of R. (So N is M's wife/partner; R is M's child.) - R × S: R is the brother of S. (So S is R's sibling — male or female; S has same parents as R, namely M and N.) - M is the father of N's child, and N is the mother of R AND S. So M is the **father of S**.

(*Note:* "M is grandfather of S" — common but WRONG. M is N's husband, not N's father. M and N are R and S's parents.)

Watch out: Students assume M+N (M father of N) means N is a child but also automatically reads "N − R" as N's child being R, leading to "M is grandfather of R, then S". The trick: N is M's WIFE, not M's daughter. Re-read the operator definitions every time.

A similar question. A − B + C × D. A is mother of B. B is father of C. C is brother of D. → A is grandmother of C; A is grandmother of D too (since C and D are siblings).

Q14. Among A, B, C and D: A is the mother of B. B is the sister of C. D is the father of A. C is daughter of whom?

What kind of problem is this? Direct family-tree build.

Solving it. Step-by-step. - A is mother of B → A → B (down 1). - B is sister of C → B and C are siblings → both have the same mother and father → A is also mother of C. - C is **daughter of A** (since C is female by phrasing "daughter").

Worth knowing: "Sister of X" implies same parents. Always check whether the question gives gender clues for the answer (here "daughter" tells us C is female).

Watch out: "Daughter of D" — D is A's father, hence C's grandfather, NOT C's father.

Q15. Pointing to a portrait, a man said, "His mother is the wife of my father's son. I have no brothers and sisters." How is the portrait related to the speaker?

What kind of problem is this? Self-referential trick.

Solving it. Step-by-step. - "My father's son" — could be the speaker himself OR a brother. But the speaker says "I have no brothers" → father's only son = the speaker. - "His mother" (i.e., portrait person's mother) = wife of father's son = wife of the speaker. - So portrait person's mother is the speaker's wife. - The speaker is the husband of portrait person's mother → the speaker is the **father of the portrait person**. - Portrait = speaker's **son** (assuming the portrait is male, as "his mother" implies).

Watch out: "Father" — students confuse direction (think portrait is the speaker's father).

A similar question. Pointing to a man, a woman said, "He is the son of the only daughter of my mother." How is he related? Only daughter of woman's mother = the woman herself. The man = woman's son.

Another twist. "She is the daughter of my grandfather's only son." Only son of grandfather = speaker's father (typically). Daughter of speaker's father = speaker's **sister**.

Section 3A — ADVANCED BLOOD RELATIONS (exam-level variations)

Q15.5 (Advanced) — Multi-clue family tree. A is B's father. B is C's brother. D is E's mother. C is married to E. F is the only child of B. How is F related to D?

Solving it. Build the tree. - A → B (B is A's child). - B and C are brothers (same parents). - D is E's mother. - C married E → E is C's spouse. - F is B's only child → F is B's child. - D is E's mother, E married C, so D is C's mother-in-law. - D is NOT directly related to B or F by blood. - F is B's child, B is brother of C, C is married to E (D's daughter). So F's uncle (C) is married to D's daughter. - F → C:

nephew/niece. C → E: husband/wife. E → D: child. So D = mother of F's uncle's wife = mother of F's aunt-by-marriage. - F is **D's daughter's husband's brother's child** — so D is grandmother-in-law of F? No, only by relation. - Most natural: D has no direct blood/marriage relation to F (F's blood relatives don't include D). - Answer: **No direct relation** (or: D is the mother-in-law of F's paternal uncle).

Q15.6 (Advanced) — Coded family with 4+ symbols. Symbols: A + B = A is mother of B; A − B = A is brother of B; A × B = A is father of B; A ÷ B = A is son of B. Decode: P + Q − R × S ÷ T.

Solving it. Step-by-step. - P + Q: P is mother of Q. - Q − R: Q is brother of R. - R × S: R is father of S. - S ÷ T: S is son of T. - So: P is mother of Q (and Q is brother of R, hence P is also mother of R). R is father of S. T is father (or mother) of S — wait, S÷T means S is son of T, so T is parent of S. But R is father of S, so T must be S's mother (only one father possible). So T is wife of R. - P (mother of R) is mother-in-law of T. - Answer: **P is mother-in-law of T**.

Section 4 — DIRECTION SENSE (5 worked + variations)

> ***Notation expansion (used throughout):*** *8-direction compass: N, NE, E, SE, S, SW, W, NW (each 45° apart). Always draw a cross (N up, E right) on rough paper before plotting.* ***Net displacement*** *= vector sum of all moves; final distance from start = √[(net N-S)² + (net E-W)²].* ***Shadow rule:*** *at sunrise (sun in east), shadows point west. At sunset (sun in west), shadows point east. At noon, no shadow (sun overhead) — equator only; in India shadows point N or S based on season.*

Q16. A walks 5 km North, then 3 km East, then 5 km South. Find his distance from the starting point.

What kind of problem is this? Net-displacement problem. Compute net N-S and net E-W, then use Pythagoras (or recognise that one component is zero).

Solving it. Step-by-step. - Set start at origin (0, 0). North is positive Y, East is positive X. - After 5 km N: position (0, 5). - After 3 km E: position (3, 5). - After 5 km S: position (3, 0). - Distance from origin = $\sqrt{3^2 + 0^2}$ = **3 km** (due east of start).

Watch out: "13 km" — students add all leg lengths (5 + 3 + 5 = 13). Wrong: those are PATH lengths, not displacement.

A similar question — All four cardinals. A walks 4 km N, 3 km E, 4 km S, 3 km W → returns exactly to start → distance = **0**.

Another twist — Diagonal walk. A walks 6 km N, then 8 km E. Distance from start = √(36 + 64) = √100 = **10 km** (a 6-8-10 Pythagorean triple).

Q17. A walks 10 km East, then turns left and walks 5 km, then turns left and walks 10 km. Find his distance from the start.

What kind of problem is this? Direction-with-turns. Track the heading after each turn.

Solving it. Step-by-step. - Initial heading: East. Walk 10 km E → position (10, 0). - Turn LEFT from East → now facing North. Walk 5 km N → position (10, 5). - Turn LEFT from North → now facing West. Walk 10 km W → position (0, 5). - Distance from origin = $\sqrt{0^2 + 5^2}$ = **5 km** (due north of start).

Worth knowing: "Left turn" rules. - Facing N → left = W; Facing E → left = N; Facing S → left = E; Facing W → left = S. - "Right turn" rules: Facing N → right = E; Facing E → right = S; Facing S → right = W; Facing W → right = N.

Watch out: "25 km" — students sum all distances (10 + 5 + 10).

A similar question — Right turns. Walk 6 km N, right turn, walk 8 km, right turn, walk 6 km → ends at (8, 0) (8 km East of start).

Q18. A man faces North. He turns 135° clockwise, then turns 180° anticlockwise. Find his final direction.

What kind of problem is this? Angular rotation tracking.

Solving it. Step-by-step. - Start: North (0°). - Turn 135° clockwise → 0° + 135° = 135° (which is SE → S, more precisely SE→S between 135° pointing → SOUTH-EAST + 45° = SOUTH-EAST? Let me be careful: 0°=N, 45°=NE, 90°=E, 135°=SE, 180°=S, 225°=SW, 270°=W, 315°=NW). So 135° clockwise from N = **SE**. - Turn 180° anticlockwise from SE: SE = 135°. Subtract 180° → −45° → +315° = **NW**.

Watch out: "SW" — students confuse clockwise vs anticlockwise after the second turn.

A similar question. Faces East, turns 90° anticlockwise, then 45° clockwise. Start E = 90°. AC 90° → 0° = N. CW 45° → 45° = **NE**.

Q19. Two friends start from the same point. A walks 12 km North; B walks 5 km East. Find the distance between them.

What kind of problem is this? Pythagoras applied to perpendicular displacements.

Solving it. Step-by-step. - A's position: (0, 12). B's position: (5, 0). - Distance AB = $\sqrt{[(5-0)^2 + (0-12)^2]}$ = $\sqrt{(25 + 144)}$ = $\sqrt{169}$ = **13 km**.

Worth knowing: Pythagorean triple 5-12-13.

Watch out: "17 km" — students add 12 + 5 linearly.

A similar question — Triple 9-40-41. A walks 9 km N, B walks 40 km E from same start. Distance = $\sqrt{(81 + 1600)} = \sqrt{1681}$ = **41 km**.

Q20. The sun rises in the east. A man's shadow at sunrise points to?

Solving it. Sun in east → light from east → shadow falls in opposite direction → **west**.

Watch out: "East" — students forget shadow is OPPOSITE to light source.

A similar question — Sunset. Sun in west at sunset → shadow points **east**.

Another twist — Time inference. A man's shadow falls to his right when he faces north. What time is it (morning or evening)? - Facing N, right side = East. Shadow east → light from west → sunset (evening).

Section 4A — ADVANCED DIRECTION (exam-level variations)

Q20.5 (Advanced) — Multi-leg path with rotation. A man starts at his home facing East. He walks 10 km, turns right and walks 6 km, turns right and walks 4 km, turns left and walks 8 km. Find his final position from home.

Solving it. Track direction + position step by step. - Start at (0, 0), facing East. - Walk 10 km E → (10, 0); facing E. - Turn right (E → S) → walk 6 km → (10, −6); facing S. - Turn right (S → W) → walk 4 km → (6, −6); facing W. - Turn left (W → S) → walk 8 km → (6, −14); facing S. - Distance from home = $\sqrt{(6^2 + 14^2)} = \sqrt{(36 + 196)} = \sqrt{232} \approx$ **15.23 km** in SE direction.

Q20.6 (Advanced) — Two persons relative direction. A walks 5 km N from origin. B starts from same origin, walks 3 km E, then turns left and walks 5 km. Find the distance between A and B.

Solving it. - A's final position: (0, 5). - B's path: (0, 0) → walk 3 km E → (3, 0) → turn left (E → N) → walk 5 km → (3, 5). - Distance AB = $\sqrt{[(3-0)^2 + (5-5)^2]} = \sqrt{9}$ = **3 km**.

Section 5 — SEATING ARRANGEMENT (5 worked + variations)

> ***Notation expansion (used throughout):*** *Always* ***draw the seat skeleton first*** *(n cells in a row, or n-position circle). Place the most-constrained person first; permissive clues last.* ***"Immediate right/left"*** *= adjacent (1 seat away).* ***"Immediately between"*** *= exactly 1 cell apart.* ***Facing rule:*** *facing centre, "left" = clockwise; facing outward, "left" = anticlockwise.*

Q21. Five friends — P, Q, R, S, T — sit in a row facing north. P is to the immediate right of Q. R is at one end. S sits between Q and T. Q is 2nd from the left. Who is to the immediate right of S?

What kind of problem is this? Linear seating with constraints. Build the row position-by-position from the most-constrained clue.

Solving it. Step-by-step. - 5 positions in a row (1 to 5), all facing north → "right" = position+1. - Most-constrained clue: Q is 2nd from left → Q at position **2**. - P is immediate right of Q → P at position **3**. - S is between Q (pos 2) and T → S has Q on one side, T on the other → if S = pos 3, conflict with P. If S between Q and T means S adjacent to BOTH, then need Q-S-T or T-S-Q consecutive. With Q at 2, possibilities: T-S-Q at positions 1-2-3 (S=2, conflicts with Q); or Q-S-T at 2-3-4 (S=3 conflicts with P). - The constraints conflict → **Question is under-determined** as stated. (In actual exam, this would be flagged or one of the constraints would be slightly different.)

Lesson. Always test if constraints are consistent BEFORE attempting to answer. If you see contradictions, the answer is likely "cannot be determined" or the question is malformed.

A similar (consistent) question. 5 friends in a row facing north. P is at extreme left. Q is to immediate right of P. R is 4th from left. S is between Q and R. T is at extreme right. → Order: P-Q-S-R-T. So who is to immediate right of Q? **S**.

Q22. Six people A, B, C, D, E, F sit around a circular table facing the centre. A is opposite D. B is to A's right. C sits between B and D. Find the arrangement.

What kind of problem is this? Circular seating. Use a 6-position circle (1 to 6 going clockwise). When facing centre, "right" goes clockwise.

Solving it. Step-by-step. - Place A at position 1 (arbitrary anchor). - A opposite D → D at position 4 (3 places away in 6-circle). - B is to A's right (clockwise from A) → B at position 2. - C between B (pos 2) and D (pos 4) → C at position 3. - Remaining E and F fill positions 5 and 6 (order indeterminate without more clues). - Going clockwise from A: A, B, C, D, [E or F], [F or E].

Watch out: Students put B at position 6 thinking "right" means anticlockwise (left of A). When facing centre, B's right is the next CLOCKWISE seat from A.

Worth knowing — Facing-outward variant. If all faced outward, B (to A's right) would be at position 6 (anticlockwise from A in our diagram). This direction-flip is the #1 trap in circular seating.

A similar question — All facing outward. Same conditions but all face outward. Then "B right of A" → B at position 6; D opposite A still at 4; "C between B and D" going around → C at 5. Final order (clockwise from A): A, F-or-E, E-or-F, D, C, B.

Q23. Eight people sit in two rows of four, facing each other. Row 1 faces south; Row 2 faces north. P is in Row 1 and is opposite W (in Row 2). Q is to the immediate right of P. R is at one end of P's row.

Find R's position.

What kind of problem is this? Two parallel rows facing each other. Track "right" carefully — depends on which row, since rows face opposite directions.

Solving it. Step-by-step. - Row 1 (faces south): seats 1-1, 1-2, 1-3, 1-4 (left to right as you stand behind looking south). Right of person in Row 1 = increasing position (1-2, 1-3, ...). - Row 2 (faces north): mirrors Row 1 directly. "Opposite" = same column. - P at some seat in Row 1; W at SAME column in Row 2. - Q is immediate right of P → Q at column (P's column + 1). - R is at one end of Row 1 → R at column 1 or 4. - Without more constraints, R could be at 1 or 4. Most likely (and consistent) answer: P at column 2, Q at column 3, R at column 4 (or column 1).

Worth knowing: "Immediate right" + "facing each other" trap. If Row 1 faces south and Row 2 faces north, a person in Row 1 sees Row 2's left as their own right (mirror reversal). Always pin down which row each "right" applies to.

Q24. Five people — A, B, C, D, E — live on a 5-floor building (1 = ground, 5 = top). A lives above B; B above C; D below E; E on the top floor. Find C's floor.

What kind of problem is this? Floor puzzle with relative-position clues.

Solving it. Step-by-step. - E on top → E = floor **5**. - D below E → D ∈ {1, 2, 3, 4}. - A > B > C (in floor terms). - A, B, C, D occupy floors 1, 2, 3, 4 (since E has floor 5). - A > B > C → C is the lowest of A, B, C → C ∈ {1, 2}. - D's exact floor — could be any of 1, 2, 3, 4. If D ≠ {A, B, C}, D takes 1 of 4 floors. - For A > B > C to fit in 3 of the 4 floors (1, 2, 3, 4), C must be 1 or 2. - If D = 4: A, B, C in {1, 2, 3} → A=3, B=2, C=1. - If D = 1: A, B, C in {2, 3, 4} → A=4, B=3, C=2 (but then E=5, D=1, A=4 — D not below E necessarily? D=1 IS below E=5 ✓). - Multiple configurations possible. Most-common answer expected: **C = 1** (the bottom of A>B>C chain when D occupies floor 4).

A similar (more determined) question. Same puzzle but adds: "D is on the 4th floor". Then D=4, E=5, and A=3, B=2, C=1 (forced).

Q25. Seven people sit in a line. P is 3rd from the left. Q is 2nd from the right. Between P and Q there are exactly 2 people. Find the position of Q.

What kind of problem is this? Linear position with a "between" constraint.

Solving it. Step-by-step. - 7 positions (1 to 7). P at position 3 (3rd from left). - Q at position 6 (2nd from right means position 7 − 2 + 1 = 6). - Between positions 3 and 6: positions 4 and 5 → 2 people ✓. - So **Q is at position 6**.

Worth knowing: "kth from left" = position k. "kth from right" in row of n = position (n − k + 1).

Watch out: Students compute "2nd from right" as position 5 (off by one — they forget the seat itself counts).

A similar question — Find total people. A is 5th from left, 11th from right. Total = 5 + 11 − 1 = **15** (subtract 1 because A is counted twice).

Section 5A — ADVANCED SEATING (exam-level variations)

Q25.5 (Advanced) — Linear with mixed facing + 1 attribute. 6 people — A, B, C, D, E, F — sit in a row. Some face North, some face South. They like different colours: red, blue, green, yellow, orange, pink. Given: - A faces North; sits at leftmost end; likes red. - F is at rightmost end; faces South. - B sits to immediate right of A; faces opposite to A. - C is third from right; likes blue. - D likes pink; sits between B and C. - E likes green. - Person at position 3 likes yellow.

Solving it. Build positions 1-6 from left. - Pos 1: A (N, red). - Pos 2: B (S, since opposite A). - Pos 6: F (S). - Pos 4: C (third from right means position 4; likes blue). - D between B and C → D at pos 3 (likes pink — but pos 3 should like yellow; contradiction unless pink ≠ yellow). - Re-check: D likes pink AND person at pos 3 likes yellow → D is NOT at pos 3. - "Between B (pos 2) and C (pos 4)" with one person between → only pos 3 works. Conflict. - This puzzle has no consistent solution as stated — flag and move on. (In real exam, double-check the question for typos.)

Q25.6 (Advanced) — Circular with attribute (Banks Mains). 8 people sit around a circular table facing the centre. They have different professions: Doctor, Engineer, Lawyer, Teacher, Singer, Dancer, Writer, Painter. Given: - A (Doctor) sits 3 seats to the right of B (Engineer). - C is opposite A. - The Lawyer sits 2 seats left of C. - D (Singer) sits between Lawyer and Doctor. - E and F sit adjacent; one is Painter.

Method. Assign positions 1-8 clockwise. Place B at pos 1; then A at pos 4 (3 right of B). C opposite A → C at pos 8. Lawyer 2 left of C → pos 6. D between Lawyer (pos 6) and Doctor (pos 4) → D at pos 5. Remaining: E, F, G, H at pos 2, 3, 7. Use "E and F adjacent" to pin down.

Method takeaway. Multi-attribute Banks-Mains puzzles need a table with rows = positions, columns = attributes; fill iteratively.

Section 6 — PUZZLES — FLOOR / SCHEDULING (5 worked + variations)

> ***Notation expansion (used throughout): Floor stacking:*** *floor 1 = ground (bottom), higher numbers = upper floors. "A above B" means A's floor number > B's floor number.* ***Days:*** *Mon, Tue, Wed, Thu, Fri, Sat, Sun. "X days after" is exclusive (not counting day 0).* ***Multi-attribute puzzle:*** *rows = persons; columns = attributes (city, sport, ...). Build a table and fill in cells from clues.*

Q26. Five boxes A, B, C, D, E are stacked vertically. A is above B. C is below D. E is at the top. B is at the bottom. Find the order.

What kind of problem is this? Box-stacking puzzle.

Solving it. Step-by-step. - E at top → position **5**. - B at bottom → position **1**. - A above B → A in positions {2, 3, 4}. - C below D → C's position < D's position. - Remaining slots 2, 3, 4 for A, C, D. - For C < D and A also in same slots: try D=4, C=3, A=2. - Verify: bottom-to-top: B (1), A (2), C (3), D (4), E (5). All conditions satisfied ✓. - Order top to bottom: **E, D, C, A, B**.

A similar question. 4 boxes stacked. P above Q. R between P and Q. S at bottom. → S(1), Q(2), R(3), P(4).

Q27. Seven people meet on different days from Monday to Sunday (one per day). P meets on Wednesday. Q meets two days after P. R meets the day after Q. On which day does R meet?

What kind of problem is this? Day-of-week puzzle with relative timing.

Solving it. Step-by-step. - P = Wednesday. - Q = Wed + 2 days = **Friday**. - R = Fri + 1 day = **Saturday**.

Watch out: Students count "two days after" inclusively — start counting from P's day instead of skipping it. "Two days after Wed" means Fri (skip Thu), not Thu.

A similar question. P meets Tuesday; Q meets 3 days after P; R meets day before Q. → Q = Friday. R = Thursday.

Q28. Five people — A, B, C, D, E — live in 5 different cities (Delhi, Mumbai, Kolkata, Chennai, Bangalore) and play 5 different sports (Cricket, Football, Tennis, Hockey, Badminton). A plays cricket and lives in Delhi. B plays football, lives in Mumbai. C lives in Kolkata, plays tennis. D plays hockey. E lives in Chennai. Where does D live?

What kind of problem is this? Multi-attribute table-fill puzzle.

Solving it. Build the table:

Person	City	Sport
A	Delhi	Cricket
B	Mumbai	Football
C	Kolkata	Tennis
D	?	Hockey
E	Chennai	?

- Cities used: Delhi, Mumbai, Kolkata, Chennai. Remaining city = **Bangalore** → D lives in Bangalore.
- Sports used: Cricket, Football, Tennis, Hockey. Remaining sport for E = **Badminton**.

A similar question — Find E's sport. From above: E plays **Badminton**.

Q29. Six friends are ranked by height. A is taller than B but shorter than C. D is taller than C. E is shorter than B. F is the shortest. Find the order from tallest to shortest.

What kind of problem is this? Linear ranking from inequality clues.

Solving it. Step-by-step. - D > C (D taller than C). - C > A > B (A taller than B, A shorter than C). - B > E (E shorter than B). - F is the shortest → F is below all. - Combine: **D > C > A > B > E > F**.

A similar question. 5 students ranked by marks. P > Q > R; S < R; T tallest. Order = T > P > Q > R > S.

Q30. Ages of three siblings: P is twice as old as Q. R is 5 years younger than P. Sum of all three ages = 50 years. Find Q's age.

What kind of problem is this? Linear age equation. Express all in terms of one variable.

Solving it. Step-by-step. - Let Q's age = x. - P = 2x (twice Q). - R = P − 5 = 2x − 5. - Sum: x + 2x + (2x − 5) = 50. - 5x − 5 = 50 → 5x = 55 → x = **11 years**. - Verify: Q = 11, P = 22, R = 17. Sum = 11 + 22 + 17 = **50** ✓.

A similar question. A is thrice as old as B. C is 4 years older than B. Sum = 30. Find B. → B = x; A = 3x; C = x+4. Sum: x + 3x + x+4 = 30 → 5x = 26 → x = 5.2 (not whole — rare in exams; usually numbers chosen to give whole).

Another twist — Average involved. Average age of A, B, C = 25; A is twice B's age; C is 5 less than B. Sum = 75. 2B + B + (B−5) = 75 → 4B = 80 → B = 20. So A = 40, C = 15. Average = 75/3 = 25 ✓.

Section 6A — ADVANCED PUZZLES (exam-level variations)

Q30.5 (Advanced) — Floor + flat + colour 3-attribute puzzle. 6 people live on 3 floors of a building, 2 flats per floor (Flat-1 East, Flat-2 West). Each person likes a different colour: Red, Blue, Green, Yellow, Orange, Pink. - A lives on top floor (Flat-1). - B lives in Flat-2 of the bottom floor; likes Red. - C lives directly above B; likes Green. - D likes Blue; lives in Flat-1 of the middle floor. - E lives in Flat-2 of an even-numbered floor; likes Pink. - F is the only remaining person.

Solving it. Build the table:

Floor	Flat-1	Flat-2
3 (top)	A	?
2 (mid)	D (Blue)	?
1 (bot)	?	B (Red)

- C directly above B → C at Floor 2, Flat-2.
- E in Flat-2 of even floor → only Floor 2 is even (1, 2, 3); Flat-2 of Floor 2 is C. Conflict.
- Re-read: maybe even-numbered floor in 1-3 means Floor 2. So E at (2, Flat-2) — but C is there. Question has conflicting clues unless interpreted differently.

Method takeaway. Multi-attribute puzzles need careful clue-by-clue placement; flag inconsistent setups.

Q30.6 (Advanced) — Day + month combo (Banks Mains). 6 people meet on different days (Mon to Sat) of different months (Jan, Feb, Mar). Each month has exactly 2 meetings. - A meets in Jan, on Mon. - B meets in Mar. - C meets day after B, same month. - D meets in Feb. - E meets day before A, same month. - F is the only person remaining.

Solving it. Build:

Month	Day 1	Day 2
Jan	E (Sun? — day before Mon)	A (Mon)
Feb	D	F
Mar	B	C (day after B)

- E meets day before Monday = Sunday. But days are Mon-Sat per problem; Sunday not allowed. Conflict — flag.
- (If days expanded to include Sun, E = Sunday in Jan.)

Section 7 — INEQUALITIES (3 worked + variations)

> ***Notation expansion (used throughout): >*** *strict greater than (cannot be equal).* ***≥*** *greater than or equal to. Same for* ***<*** *and* ***≤****.* ***=*** *equal.* ***Chain rule:*** *A > B and B > C → A > C. But A > B and B = C → A > C (strict carries through). And A ≥ B and B > C → A > C (strict wins over ≥).* ***Inequality "breaks":*** *A > B and B < C tells us NOTHING about A vs C (the chain reverses direction at B).*

Q31. Statements: A > B ≥ C; D ≤ B. Conclusions: (I) A > D, (II) D < C. Which conclusion(s) follows?

What kind of problem is this? Inequality chain analysis. Trace each conclusion through the given chain.

Solving it. Step-by-step. - Given: $A > B$; $B \geq C$; $D \leq B$.

Conclusion I: A > D? - We know $A > B$ and $D \leq B$ → so $D \leq B < A$ → **D < A** is true → so **A > D** ✓.

Conclusion II: D < C? - $D \leq B$ and $B \geq C$ → $D \leq B$ and $C \leq B$. This tells us both D and C are below or equal to B, but does NOT order D vs C. - D could be less than, equal to, or greater than C. **Conclusion II does NOT follow strictly.**

Answer: Only I follows.

Worth knowing: Strict-vs-equality logic. - $>$ beats $\geq$ (i.e., $A > B \geq C$ → $A > C$ strict). - $<$ beats $\leq$ (i.e., $A < B \leq C$ → $A < C$ strict). - $\geq \geq = \geq$ (cannot conclude strict).

Watch out: Students conclude both follow because both "look" derivable. Always TRACE the chain — if there's a "≤" or "≥" you cannot turn into strict, conclusion is uncertain.

A similar question — Both follow. $A > B > C \geq D$. Conclusions: (I) $A > D$ (II) $B > D$. Both follow ✓ (chain doesn't break).

Q32. Statements: $P > Q = R \geq S$. Conclusions: (I) $P > S$, (II) $Q \geq S$.

Solving it. - $P > Q \geq S$ → $P > S$ (chain holds, $>$ carries through $\geq$) ✓. - $Q = R$ and $R \geq S$ → $Q \geq S$ ✓. - **Both conclusions follow.**

Q33. Statements: $A \geq B = C$; $D > A$. Conclusions: (I) $D > C$, (II) $A \geq C$.

Solving it. - $D > A$ and $A \geq C$ → $D > A \geq C$ → $D > C$ ✓. - $A \geq B$ and $B = C$ → $A \geq C$ ✓. - **Both conclusions follow.**

Section 7A — ADVANCED INEQUALITIES (exam-level variations)

Q33.5 (Advanced) — Coded inequality with 2 statements + 4 conclusions. Symbols: @ = >; # = ≥; & = <; $ = ≤; * = =. **Statements:** P @ Q # R; S ∗ T & R. **Conclusions:** (I) P @ T (II) S & P (III) Q & T (IV) R @ S.

Solving it. Decode and analyse. - Decoded: $P > Q \geq R$; $S = T < R$. - (I) $P > T$? $P > Q \geq R > T$ → $P > T$ ✓. - (II) $S < P$? $S = T < R \leq Q < P$ → $S < P$ ✓. - (III) $Q < T$? $Q \geq R > T$ → $Q > T$ (not $Q < T$). ✗. - (IV) $R > S$? $S = T < R$ → $R > S$ ✓. - **Conclusions I, II, IV follow; III does not.**

Q33.6 (Advanced) — Either/or conclusion. Statements: A > B = C; D > B. Conclusions: (I) A > D (II) D > A. **Solving it.** A > B and D > B — both bigger than B, but no order between A and D. So either I or II MUST be true (one is greater than the other; cannot both fail). **Either I or II follows.**

Section 8 — SYLLOGISM (5 worked + variations)

> ***Notation expansion (used throughout): All A are B*** *= every A is B.* ***Some A are B*** *= at least one A is B.* ***No A is B*** *= no overlap.* ***Conversion rules:*** *"Some A are B" ⇔ "Some B are A" (always converts). "No A is B" ⇔ "No B is A" (always converts). "All A are B" does NOT convert — only "Some B are A" follows.* ***Universal method:*** *Draw Venn diagrams. Test each conclusion against ALL valid Venn arrangements.*

Q34. Statements: All cats are dogs. Some dogs are cows. Conclusions: (I) Some cats are cows. (II) Some cows are dogs.

Solving it.

Step 1 — Visualise. - All cats are dogs → cats circle is INSIDE dogs circle. - Some dogs are cows → dogs circle and cows circle OVERLAP (somewhere).

Step 2 — Test Conclusion I. "Some cats are cows." - The cows-overlap with dogs could be anywhere — it might or might not include the cats subset. - We cannot conclude that any cats are cows. → **Conclusion I does NOT follow**.

Step 3 — Test Conclusion II. "Some cows are dogs." - This is the direct conversion of "Some dogs are cows" (which always holds for "some" statements). - → **Conclusion II follows**.

Answer: Only II follows.

Watch out: Students assume "if cats ⊂ dogs, and some dogs ∩ cows, then some cats ∩ cows." WRONG — the cow overlap might be entirely OUTSIDE the cats subset.

Q35. Statements: No book is pen. All pens are pencils. Conclusions: (I) Some pencils are not books. (II) No pencil is book.

Solving it. - No book is pen → books and pens DO NOT overlap. - All pens are pencils → pens ⊂ pencils. - So pens (= part of pencils) is disjoint from books → that part of pencils that overlaps with pens is definitely not books → **at least some pencils are not books** ✓. - Conclusion II ("No pencil is book") — too strong. There MIGHT be other pencils (not pens) that overlap with books, since we have no info about non-pen pencils vs books. → does NOT follow strictly. - **Only I follows.**

Q36. Statements: Some doctors are professors. All professors are teachers. Conclusions: (I) Some doctors are teachers. (II) Some teachers are doctors.

Solving it. - Some doctors are professors → doctors ∩ professors ≠ empty. - All professors are teachers → professors ⊂ teachers. - So doctors ∩ professors ⊂ teachers. Hence those doctors are teachers → **Some doctors are teachers** ✓. - By conversion of "Some doctors are teachers" → **Some teachers are doctors** ✓. - **Both follow.**

Q37. Statements: All A are B. All B are C. Conclusions: (I) All A are C. (II) Some C are A.

Solving it. - A ⊂ B and B ⊂ C → A ⊂ C → **All A are C** ✓. - By conversion of "All A are C" → **Some C are A** ✓. - **Both follow.**

Q38. Statements: No man is animal. Some animals are pets. Conclusion: Some pets are not men.

Solving it. - Some animals are pets → there exist some entities that are both animals AND pets. - These animal-pets are NOT men (since "no man is animal"). - So **at least some pets are not men** ✓. **Conclusion follows.**

Section 8A — ADVANCED SYLLOGISM (exam-level variations)

Q38.5 (Advanced) — 3 statements + 3 conclusions. Statements: All cars are vehicles. No vehicle is a fruit. Some fruits are red. Conclusions: (I) No car is a fruit. (II) Some red things are not cars. (III) All cars are not fruits.

Solving it. - Cars ⊂ Vehicles; Vehicles ∩ Fruits = ∅; Some Fruits ∩ Red. - (I) Cars are vehicles; no vehicle is fruit → no car is fruit ✓. - (II) Some fruits are red. Those red fruits are NOT cars (since fruits and cars are disjoint via vehicles). So some red things (= those fruit-reds) are not cars ✓. - (III) "All cars are not fruits" = "No cars are fruits" = same as I ✓. - **All three follow.**

Q38.6 (Advanced) — Possibility conclusion (Banks Mains). Statements: Some pens are pencils. All pencils are erasers. Conclusions: (I) All pens being erasers is a possibility. (II) Some erasers are not pens.

Solving it. - (I) Pens overlap with pencils. Pencils are inside erasers. The portion of pens NOT in pencils could ALSO be inside erasers (no statement forbids it). So "all pens being erasers" is POSSIBLE. ✓. - (II) Erasers contain all pencils PLUS possibly more. The "more" portion may or may not include the non-pen pens. Cannot conclude DEFINITELY some erasers are not pens. (Could be all erasers ARE pens in one valid Venn.) ✗. - **Only I follows.**

Memory peg: "Possibility" follows whenever NO statement directly forbids the proposed Venn arrangement.

Section 9 — INPUT-OUTPUT (3 worked + variations)

> ***Notation expansion (used throughout):*** *Banks Mains specialty. A "machine" applies a fixed rule to an input list, producing successive steps. The candidate must IDENTIFY the rule from input → step 1, then predict later steps.* ***Universal first move:*** *Compare INPUT with STEP 1 element-by-element. The rule will be visible in 30 seconds with practice.*

Q39. Input: 12 24 6 18 9. Step 1: 6 12 24 18 9. Find the rule.

Solving it. Compare position-by-position. - Position 1: 12 → 6. Position 2: 24 → 12. Position 3: 6 → 24. Position 4-5: unchanged (18, 9). - The smallest number (6) moved from position 3 to position 1; the rest of the original first 3 positions shifted right by one. - **Rule:** Move the smallest of the first three numbers to position 1; shift the others right.

Watch out: "Sorted" — students assume full alphabetical/numerical sort, but only the smallest moved.

A similar question — Smallest to leftmost (whole list). Input: 5 9 3 7 1. Step 1: 1 5 9 3 7. → Rule: smallest moved to position 1; rest shift right.

Q40. Input: cat dog ant bat. Step 1: ant bat cat dog. Find the rule.

Solving it. Compare. The 4 words are now in alphabetical order. **Rule: Sort alphabetically (ascending)**.

Q41. Input: 5 9 3 7 11. Step 1: 5 9 3 7 121. Step 2: 5 9 3 49 121. Find the next steps.

Solving it. Step-by-step. - Step 1: position 5 (which was 11) → 121 = 11^2. Other positions unchanged. - Step 2: position 4 (which was 7) → 49 = 7^2. Position 5 stays at 121. - **Rule:** In each step, square the rightmost-not-yet-squared number. - Step 3: position 3 (3) → 9. Step 3 result: 5 9 9 49 121. - Step 4: position 2 (9) → 81. Step 4 result: 5 81 9 49 121. - Step 5: position 1 (5) → 25. Step 5 result: 25 81 9 49 121. - After step 5, all positions squared.

Section 9A — ADVANCED INPUT-OUTPUT (exam-level variations)

Q41.5 (Advanced) — 2-rule machine. Input: 24 17 35 12 48 9. Step 1: 9 24 17 35 12 48. Step 2: 9 12 24 17 35 48.

Find the rule and Step 3.

Solving it. - Step 1: smallest (9) moved to position 1; rest shifted right preserving order. - Step 2: next-smallest of remaining (12) moved to position 2; rest shifted right. - **Rule:** Each step, find the smallest unsorted element and place it in next-leftmost unsorted position; rest shift right. - Step 3: Smallest unsorted = 17; move to position 3 → 9 12 17 24 35 48 (already sorted).

Q41.6 (Advanced) — Symbol-shift rule. Input: A1 B5 C3 D7 E2. Step 1: B5 A1 C3 D7 E2 (largest letter+number combo moved to front).

Method. Each step identifies max-element by sum or by criterion; tracking which position becomes target.

Section 10 — DATA SUFFICIENCY (3 worked + variations)

> ***Notation expansion (used throughout):*** *Question + 2 statements (I, II). Decide which subset of statements is sufficient to solve.* ***Standard answer choices:*** *(a) Statement I alone is sufficient. (b) Statement II alone is sufficient. (c) Each statement ALONE is sufficient. (d) BOTH statements together are needed (neither alone suffices). (e) Neither alone nor together is sufficient.* ***Discipline:*** *test each statement IN ISOLATION. Do not look at one while testing the other.*

Q42. Find the value of x. Statement I: x + 5 = 12. Statement II: x − 3 = 4.

Solving it. - I alone: x = 12 − 5 = **7** → I sufficient. - II alone: x = 4 + 3 = **7** → II sufficient. - **Each alone is sufficient. Answer: (c).**

Q43. Is x an even number? Statement I: x = 4y, where y is a positive integer. Statement II: x is divisible by 3.

Solving it. - I alone: 4y = 4 × (positive integer) = always EVEN (multiple of 4 is always multiple of 2). → Yes, x is even. **I sufficient.** - II alone: divisible by 3 → x could be 3 (odd), 6 (even), 9 (odd), 12 (even), … → cannot decide parity. **II not sufficient.** - **Only I alone is sufficient. Answer: (a).**

Q44. Find the perimeter of a rectangle. Statement I: Length = 10 cm. Statement II: Area = 30 cm².

Solving it. - I alone: Length given but no width → cannot find perimeter. **I not sufficient.** - II alone: Area 30 = L × W → infinite (L, W) pairs (1 × 30, 2 × 15, 3 × 10, ...) → cannot find perimeter. **II not sufficient.** - I + II together: L = 10, Area = 30 → W = 30 / 10 = 3 → Perimeter = 2(10 + 3) = **26 cm**. ✓ - **Both together are needed. Answer: (d).**

Section 10A — ADVANCED DATA SUFFICIENCY (exam-level variations)

Q44.5 (Advanced) — 3-statement DS. Find the age of A. (I) A is 5 years older than B. (II) B is 25 years old. (III) A + B = 55.

Solving it. - I + II → A = B + 5 = 30. Sufficient. - I + III → 2B + 5 = 55 → B = 25, A = 30. Sufficient. - II + III → A = 55 − 25 = 30. Sufficient. - **Any TWO statements together suffice. Single statement alone does NOT.**

Q44.6 (Advanced) — Quant DS with conditions. Is x positive? (I) $x^2 > 4$. (II) $x^3 > 0$.

Solving it. - I: $x^2 > 4 \rightarrow x > 2$ or $x < -2$. Could be positive OR negative. Insufficient alone. - II: $x^3 > 0 \rightarrow x > 0$. Sufficient alone. - **Only II alone suffices.**

Section 11 — NON-VERBAL (3 worked + variations)

> ***Notation expansion (used throughout): Mirror image (vertical axis)*** *= reflection across a vertical line; left-right swap. Reverse the order of letters AND mirror each letter glyph.* ***Water image (horizontal axis)*** *= reflection across a horizontal line; top-bottom swap.* ***Painted cube (n × n × n cut into 1 cm³ small cubes):*** *Corners = 8 (3 faces painted); Edges = 12(n−2) (2 faces painted); Face-centres = 6(n−2)² (1 face painted); Interior = (n−2)³ (0 faces painted).*

Q45. Find the mirror image of the word INDIA along a vertical axis.

What kind of problem is this? Mirror image — vertical-axis reflection.

Solving it. Step-by-step. - **Step 1 — Reverse the order of letters.** INDIA → AIDNI. - **Step 2 — Mirror each glyph.** Some letters look the same in mirror (A, H, I, M, O, T, U, V, W, X, Y); others flip visibly (B, C, D, E, K, etc.). - For INDIA: A → A; I → I; D → ᗡ (D-reversed); N → И (N-reversed); I → I. - Final visible answer: **A I ᗡ И I** (read left-to-right as AIDNI with D and N glyph-flipped).

Worth knowing: Mirror-symmetric letters (no glyph change): **A, H, I, M, O, T, U, V, W, X, Y**.

Watch out: Students just reverse letter order without flipping the individual glyphs.

A similar question — Water image of "BOAT". Vertical flip of each letter. B mostly looks the same (top-bottom roughly symmetric); O symmetric; A flips upside-down (∀); T inverts to ⊥. Result: B O ∀ ⊥.

Q46. A cube of side 4 cm is painted on all 6 faces, then cut into 1 cm^3 small cubes. How many small cubes have exactly 1 face painted?

What kind of problem is this? Painted-cube counting. Use the standard formula.

Solving it. Step-by-step. - n = 4 (cube cut 4 × 4 × 4 = 64 small cubes). - Face-centre small cubes (1 face painted) = **$6 \times (n - 2)^2 = 6 \times (4 - 2)^2 = 6 \times 4 = 24$**.

Worth knowing: Memorise the painted-cube formulas cold.

Type	Count for n × n × n cube
Corner (3 faces painted)	always 8
Edge (2 faces painted)	$12(n - 2)$
Face-centre (1 face painted)	$6(n - 2)^2$
Interior (0 faces painted)	$(n - 2)^3$
Total	n^3

- Verify for n = 4: $8 + 12{\times}2 + 6{\times}4 + 2^3 = 8 + 24 + 24 + 8 =$ **64** $= 4^3$ ✓.

A similar question — Cube n=5, find 0-painted. Interior = $(5-2)^3 = 27$.

Another twist — n=6, find 2-painted. Edge = 12 × 4 = 48.

Q47. A large triangle is divided into 4 smaller triangles by connecting the midpoints of its three sides. How many triangles are there in the figure?

Solving it. - 4 small triangles + 1 large outer triangle = 5? But also the central inverted triangle counts. - Counting: 3 corner triangles + 1 inverted middle + 1 large outer = **5 triangles total**.

Worth knowing: For figure-counting problems, count systematically by SIZE — smallest first, then medium, then largest. Sum.

Section 11A — ADVANCED NON-VERBAL (exam-level variations)

Q47.5 (Advanced) — Paper folding. A square paper is folded once along its diagonal, then a small triangular notch is cut at the centre. When unfolded, what shape do the cuts form?

Solving it. - Fold along diagonal → triangular shape. - Cut a triangle at centre → on unfolding, the cut REFLECTS across the diagonal. - Original cut + reflection = a 4-sided shape (rhombus or kite, depending on cut location). - **Result:** rhombus / kite shape at the centre of the unfolded paper.

Q47.6 (Advanced) — Cube assembly from net. A cube net has 6 squares labelled 1-6 in a cross arrangement: 1 (top), 2 (middle of arm), 3 (right of 2), 4 (left of 2), 5 (bottom of 2), 6 (below 5). When folded, which face is opposite face 1?

Solving it. - In a cross-net layout: centre face (2) becomes one face of the cube. - Faces 1 and 5 (top and bottom of the central column) become OPPOSITE. - Faces 3 and 4 (right and left of centre) become OPPOSITE. - Face 6 (below 5) becomes opposite to face 2 (centre). - **So opposite of face 1 = face 5.**

Section 12 — RANKING + ALPHABET (3 worked + variations)

> ***Notation expansion (used throughout): kth from left*** *in a row of n = position k.* ***kth from right*** *= position (n − k + 1).* ***Total when both ranks given:*** *n = (left-rank) + (right-rank) − 1 (subtract 1 because the person is counted in both).* ***After moving:*** *if A's left-rank changes from k to k', A moved (k' − k) positions to the right.*

Q48. A is 7th from the left and 12th from the right in a row. Find the total number of people in the row.

Solving it. - Total = (left rank) + (right rank) − 1 = 7 + 12 − 1 = **18**.

Watch out: "19" — students add without subtracting 1, double-counting A.

A similar question. B is 5th from left and 8th from right. Total = 5 + 8 − 1 = **12**.

Another twist — Find rank of someone else. Same row of 18, A is at position 7. C is 4 positions to the right of A → C at position 11. C's rank from right = 18 − 11 + 1 = **8**.

Q49. Arrange the words alphabetically: SUN, MOON, STAR, MARS, EARTH.

Solving it. Compare letter-by-letter. - E (EARTH) < M (MARS, MOON) < S (STAR, SUN). - Within M-words: MARS (M-A-R-S) vs MOON (M-O-O-N). A < O → MARS first. - Within S-words: STAR (S-T-A-R) vs SUN (S-U-N). T < U → STAR first. - Final order: **EARTH, MARS, MOON, STAR, SUN**.

Q50. Find the positions of the letter P in the word "PHILOSOPHY".

Solving it. - Letters: P-H-I-L-O-S-O-P-H-Y (10 letters total). - P appears at position 1 and position 8 → **positions 1 and 8**.

Section 12A — ADVANCED RANKING (exam-level variations)

Q50.5 (Advanced) — Position change after movement. In a row of 25 students, A is at the 8th position from the left. He moves 5 places to the right. What is his new position from the right?

Solving it. - Original from left = 8 → from right = 25 − 8 + 1 = 18. - After moving 5 right, new position from left = 13. - New position from right = 25 − 13 + 1 = **13**.

Q50.6 (Advanced) — Queue with two reference points. In a queue, B is at the 7th position from the front. C is 4 places ahead of B. If C is 12th from the back, find total people in the queue.

Solving it. - B at position 7 from front. C is 4 ahead of B → C at position 7 − 4 = 3 from front. - C is also 12th from back. So total = 3 + 12 − 1 = **14 people**.

Section 13 — ADVANCED REASONING TYPES (Banks Mains 2024-26 patterns)

> ***Why this section exists.*** *Modern Banks PO Mains and SBI PO use NEW reasoning patterns: floor + flat + colour multi-attribute puzzles, machine input-output with complex rules, coded inequality with 3 statements, advanced syllogism with possibility ("at least", "all may be"), seating with 8+ people in 3-row layouts. This section covers each.*

Q51 (Advanced) — Floor + flat + attribute puzzle (Banks Mains 5-Q set).

Setup: 8 people — A, B, C, D, E, F, G, H — live on 4 floors of a building (floor 1 = ground, floor 4 = top). Each floor has 2 flats: Flat-1 (East) and Flat-2 (West). - A lives on floor 4, Flat-1. - B lives on floor 1, Flat-2. - C lives directly above E (same flat number). - D lives in Flat-2 of an even-numbered floor. - F lives in Flat-1 of floor 2. - G lives on floor 3. - H is the only person remaining.

Sub-Q: On which floor and flat does H live?

Solving it. Build the table.

Floor	Flat-1 (East)	Flat-2 (West)
4	**A**	?

Floor	Flat-1 (East)	Flat-2 (West)
3	?	?
2	**F**	?
1	?	**B**

- D in Flat-2 of even floor → D at Flat-2 of floor 4 OR floor 2.
- C above E (same flat) → pairs: (C, E) at floors (4, 3), (3, 2), (2, 1).
- G on floor 3.
- A is at (4, Flat-1) and F at (2, Flat-1). For "C above E in same flat", look at remaining slots: E could be at (3, Flat-1) and C at (4, Flat-1) — but A is there. Try E at (3, Flat-2) and C at (4, Flat-2). G is on floor 3 — could be at (3, Flat-1).
- So: floor 3 Flat-1 = G; floor 3 Flat-2 = E; floor 4 Flat-2 = C.
- D in Flat-2 of even floor → only (2, Flat-2) is left → D at (2, Flat-2).
- Remaining slot = (1, Flat-1) → H at floor 1, Flat-1.

Answer: H lives on floor 1, Flat-1.

Method takeaway. Always build a table; lock most-constrained clues first; eliminate by exhaustion.

Q52 (Advanced) — Machine Input-Output with 2-rule shift.

Input: 25 38 14 47 19 56. **Step 1:** 14 25 38 19 47 56. **Step 2:** 14 19 25 38 47 56.

Find the rule and Step 3 + Step 4.

Solving it. - Compare Input → Step 1: 14 (smallest unsorted-prefix item) moved to position 1. - Compare Step 1 → Step 2: 19 (next-smallest) moved into position 2 of the sorted prefix. - **Rule:** Sort ascending one position at a time, sliding next-smallest into place. - Step 2 already has the first 4 sorted: 14, 19, 25, 38. Remaining 47, 56 — already in order. - Step 3 (and final): no further changes needed since all sorted. **Final sorted output:** 14 19 25 38 47 56.

A more complex variant. Input: 28 14 35 21 49 7 56. Each step finds smallest of unsorted suffix and moves it into the sorted prefix.

Q53 (Advanced) — Coded inequality with 3 statements.

Symbols: @ means >; # means ≥; & means <; $ means ≤; * means =. **Statements:** P @ Q # R; S * T & R; U $ S. **Conclusions:** (I) P > S (II) U < R (III) T < P

Find which conclusions follow.

Solving it. Decode and analyse. - P > Q ≥ R; S = T < R; U ≤ S.

Conclusion I: $P > S$? - $P > Q \geq R > T = S \rightarrow P > S$ ✓.

Conclusion II: $U < R$? - $U \leq S = T < R \rightarrow U \leq S < R \rightarrow U < R$ ✓.

Conclusion III: $T < P$? - $T = S$; from above, $P > S \rightarrow P > T \rightarrow T < P$ ✓.

All three conclusions follow.

Q54 (Advanced) — Syllogism with "possibility" conclusion.

Statements: Some pens are pencils. All pencils are erasers. **Conclusions:** (I) All pens being erasers is a possibility. (II) Some erasers are definitely pens.

Solving it. - Some pens ∩ pencils → those overlap-pens are also erasers (since all pencils are erasers). - So at least some pens ARE erasers definitely. → **Some pens are erasers** is true. - (I) "All pens being erasers is a POSSIBILITY" — the pens NOT in pencils-overlap could also be in erasers (no statement prohibits this). Diagram allows it. → **Possibility holds** ✓. - (II) "Some erasers are definitely pens" = converse of "some pens are erasers", which holds. ✓. - **Both follow.**

Worth knowing: "Possibility" conclusions hold whenever NO statement contradicts the proposed Venn arrangement.

Q55 (Advanced) — Direction with displacement + rotation combo.

A man starts walking facing North. He walks 5 km, then turns 45° clockwise and walks $7\sqrt{2}$ km, then turns 90° anticlockwise and walks 3 km. Find his final position relative to start.

Solving it. - Start: facing N, position (0, 0). - Walk 5 km N → position (0, 5). - Turn 45° clockwise → now facing NE. Walk $7\sqrt{2}$ km in NE → moves equally in N and E: $7\sqrt{2} \times \cos 45° = 7$ in N, 7 in E. Position: (7, 12). - Turn 90° anticlockwise from NE → now facing NW. Walk 3 km NW: $-3 \cos 45° = -3/\sqrt{2} \approx -2.12$ in E, $+3/\sqrt{2} \approx +2.12$ in N. Position: $(7 - 2.12, 12 + 2.12) = (4.88, 14.12)$. - Distance from start $= \sqrt{(4.88^2 + 14.12^2)} \approx \sqrt{(23.8 + 199.4)} = \sqrt{223.2} \approx$ **14.94 km**, direction ≈ NNE.

Q56 (Advanced) — 2-row seating with 8 people, mixed facing.

Setup: 8 people — A, B, C, D in Row 1 (facing south); E, F, G, H in Row 2 (facing north). Persons in Row 1 face persons in Row 2 directly. - B is opposite F. - A is to immediate left of B. - C is at one end of Row 1. - E is at one end of Row 2. - G is opposite C. - D is between A and C.

Find the arrangement.

Solving it. Build two rows of 4 columns.

Row 1 (faces S)	col 1	col 2	col 3	col 4
Row 2 (faces N)	col 1	col 2	col 3	col 4

- B opposite F → same column.
- A immediate left of B (Row 1 facing south → "left" is from B's perspective looking south = our visual right = column+1... wait).
- For Row 1 facing south, the spectator behind sees them looking south. Person in Row 1 at column 1 has their "left" = column 2 (spectator's right). So "A immediate left of B" = A at column (B's column + 1).
- C at one end → col 1 or col 4.
- D between A and C.

This kind of multi-clue puzzle requires backtracking. Try B at col 2: A at col 3 (immediate left of B). Then for D to be "between A (col 3) and C", C must be at col 4 (since col 1 would need D at col 2, which is B's slot). D between col 3 (A) and col 4 (C) → D between... impossible since adjacent. So C at col 1, D at col 2... but B is at col 2. Try B at col 3: A at col 4. C at col 1. D between A (col 4) and C (col 1) → D at col 2. Check: F opposite B → F at Row 2 col 3. G opposite C → G at Row 2 col 1. E at one end of Row 2 → col 1 or col 4. G is at col 1, so E at col 4. H takes remaining col 2.

	col 1	col 2	col 3	col 4
Row 1 (faces S)	C	D	B	A
Row 2 (faces N)	G	H	F	E

Method takeaway. Multi-row + multi-attribute puzzles need patient table-fill, often backtracking on initial guesses.

PART E — TIMED MINI-MOCK (25 Q · 25 min)

Instructions: One mark per question, no negative marking in this self-test. Attempt all 25 within 25 minutes.

1. Number series: 5, 11, 23, 47, __ ? (a) 91 (b) 95 (c) 99 (d) 103
2. If A = 1, B = 2, ... Z = 26, the letter-value of CAR is: (a) 22 (b) 24 (c) 25 (d) 26
3. Pointing to a woman, a man says: "Her son's father is my father." How is the woman related to the man? (a) Mother (b) Sister (c) Wife (d) Daughter
4. A walks 5 km East, then 5 km North, then 5 km West. His straight-line distance from the starting point is: (a) 5 km (b) 10 km (c) 7 km (d) 0 km
5. Five persons P, Q, R, S, T sit in a row. P sits at the left end. T sits at the right end. R sits between S and T. Q is adjacent to P. Which position (from left) does S occupy? (a) 2nd (b) 3rd (c) 4th (d) 5th
6. All doctors are engineers. All engineers are scientists. Which conclusion(s) definitely follow? I — All doctors are scientists. II — Some scientists are engineers. (a) Only I (b) Only II (c) Both I and II (d) Neither
7. $P > Q > R \leq S$. Which conclusion definitely follows? I — $P > R$. II — $S > P$. (a) Only I (b) Only II (c) Both (d) Neither
8. Letter series: A, E, I, M, Q, __ ? (a) S (b) T (c) U (d) V
9. A cube of side 4 cm is painted red on all faces and then cut into 1 cm^3 small cubes. How many small cubes have exactly 2 faces painted? (a) 8 (b) 16 (c) 24 (d) 32
10. The day before yesterday was Thursday. What day is tomorrow? (a) Saturday (b) Sunday (c) Monday (d) Friday
11. Five words arranged in alphabetical order: ORANGE, MANGO, APPLE, BANANA, CHERRY. Which word comes third? (a) CHERRY (b) MANGO (c) BANANA (d) ORANGE
12. In a code, BIRD = 33 and FISH = 42 (sum of letter positions, A = 1). What is the value of CODE? (a) 25 (b) 27 (c) 29 (d) 32
13. A boat covers 24 km downstream in 3 hours and 24 km upstream in 6 hours. What is the speed of the boat in still water? (a) 4 km/h (b) 5 km/h (c) 6 km/h (d) 8 km/h
14. P is the uncle of Q. R is the sister of P. How is R related to Q? (a) Aunt (b) Sister (c) Niece (d) Cousin

15. Total number of squares of all sizes on a standard 8 × 8 chessboard: (a) 64 (b) 128 (c) 196 (d) 204
16. Number series: 8, 27, 64, 125, __ ? (a) 196 (b) 216 (c) 169 (d) 256
17. If '+' means '×', '−' means '+', '×' means '÷', '÷' means '−', then: 6 + 4 ÷ 2 × 1 − 5 = ? (a) 26 (b) 25 (c) 27 (d) 24
18. Odd one out: 16, 36, 64, 81, 100 (a) 16 (b) 36 (c) 64 (d) 81
19. Letter-pair series: AZ, BY, CX, DW, __ ? (a) EV (b) EU (c) FV (d) EW
20. Sum of all natural numbers from 1 to 100: (a) 5050 (b) 5000 (c) 5500 (d) 5100
21. A tells B: "Your mother's husband's sister is my mother." How is A related to B? (a) Brother (b) Cousin (c) Uncle (d) Nephew
22. A man walks 5 km South, then 12 km East. His straight-line distance from the starting point is: (a) 13 km (b) 17 km (c) 7 km (d) 60 km
23. If 7 days ago was Tuesday, what day is today? (a) Monday (b) Tuesday (c) Wednesday (d) Thursday
24. Letter series: A, C, F, J, O, __ ? (a) T (b) U (c) V (d) S
25. In a class, Rohan's rank from the top is 15th and his rank from the bottom is 21st. How many students are in the class? (a) 34 (b) 35 (c) 36 (d) 37

Answer key: 1-b | 2-a | 3-a | 4-a | 5-b | 6-c | 7-a | 8-c | 9-c | 10-b | 11-a | 12-b | 13-c | 14-a | 15-d | 16-b | 17-c | 18-d | 19-a | 20-a | 21-b | 22-a | 23-b | 24-b | 25-b

Quick working for tricky questions: - Q2: C=3, A=1, R=18 → 3+1+18 = **22** → (a) - Q3: "Her son's father" = her husband = speaker's father → she is the speaker's **mother** → (a) - Q7: P > Q > R means P > R is definite. S ≥ R but S vs P is unknown → only I follows → (a) - Q9: 2-face cubes = 12(n−2) = 12 × 2 = **24** → (c) - Q10: DBY = Thu → yesterday = Fri → today = Sat → tomorrow = **Sun** → (b) - Q12: BIRD = 2+9+18+4 = 33 ✓. CODE = 3+15+4+5 = **27** → (b) - Q13: Downstream = 24÷3 = 8 km/h. Upstream = 24÷6 = 4 km/h. Still water = (8+4)÷2 = **6** → (c) - Q17: Substitute ops: 6×4 − 2÷1 + 5. BODMAS: 24 − 2 + 5 = **27** → (c) - Q21: "Your mother's husband" = B's father. B's father's sister = B's aunt = A's mother → A is B's **cousin** → (b) - Q25: Total = 15 + 21 − 1 = **35** → (b)

Scoring: 23–25 = Excellent | 18–22 = Good | Below 18 = revise weak topics before your mock series.

⚠ EXAM ALERT

∗∗Error-log rule:** For every question you got wrong, write one sentence explaining what you missed. Retest yourself on only those questions 3 days later — without looking at the solution. This single habit doubles what the mock teaches you.

PART F — TRAP-RECOGNITION CARDS

Trap pattern	Where	How to spot
"left/right" facing centre vs facing outward	Circular seating	Always note facing direction; left/right flips
"Some" → assumed "All"	Syllogism	"Some" never implies "all"; conversion of I-statement is also "some"
Adding distances when net displacement asked	Direction	Always vector-add components
"between X and Y" needs 2 people OR adjacency	Seating	Read carefully; if "exactly 2 between", positions fixed
Inequality chain breaks at sign reversal	Coded inequality	A > B < C tells you nothing about A vs C
Coding rule fits one example but not other	Coding-decoding	Test rule on BOTH given codes before applying
Cube — corner vs edge vs face count	Painted cube	Memorise: corners 8, edges $12(n-2)$, faces $6(n-2)^2$, interior $(n-2)^3$
Series Δ-of-Δ not constant	Number series	Try ratios, then square/cube/Fibonacci
Mirror vs water image	Non-verbal	Mirror = vertical axis (reverse order); Water = horizontal axis (flip vertically)
"Nephew/niece vs grandson"	Blood relations	Always draw the family tree first

Drill all 50 + the mini-mock thrice for >90% mastery. Then attempt 4-5 PYQ papers from the past 3 years to lock in real exam-pace.

PART G — ADVANCED BANKS-MAINS PUZZLES (30 worked examples)

Banks Mains gives 4-5 high-difficulty puzzle sets, each carrying 5 marks. Mastering these 30 archetypes covers 95% of what you'll see.

G.1 LINEAR SEATING — single row, mixed facing (5)

P1. Eight people A, B, C, D, E, F, G, H sit in a row, some facing N, some facing S. A is at one end. B sits 3rd to the right of A. C sits 2nd to the left of B. D sits adjacent to C. E sits 2nd to the right of B. F sits between E and D. G is at the other end facing N. H faces opposite to G. Find arrangement.

Method (compressed). Write 8 positions. Place A at pos 1. B at 4 (3rd to right of A). C at 2 (2nd to left of B). D adj to C → D=1 or 3, but A=1 → D=3. E at 6 (2nd right of B). F between E and D → F=5 (between 3 and 6, but D and E aren't adjacent, so F at 5 makes E-F-(empty)-D ordering; actual: F between E (6) and D (3) means F is in between positions, pos 5 satisfies). Remaining 7, 8 for G, H. G at end (8) facing N. H at 7 facing S. Valid arrangement found.

Watch out: "Facing direction" determines what "right" means; clarify before placing.

P2. Linear with reverse-direction clues — e.g., "Q is opposite to person at 3rd position from R" — always pivot off a known fixed person.

P3. Clue: "between A and B, there are exactly 3 people" — gives you a window of 5 positions; combine with ranking.

P4. "C is to the left of D" alone is ambiguous; need "immediate left" or "between X and Y".

P5. Always finalise positions before answering questions; the same arrangement gives all 5 answers.

G.2 LINEAR — two parallel rows facing each other (5)

P6. Five in row 1 face S; five in row 2 face N. P (row 1) sits opposite Q (row 2). R (row 1) is to the right of P. S (row 2) is opposite R. T (row 2) sits at one end. Find arrangement.

Method. Draw two rows of 5 each, facing each other. "Opposite" means same column index. Place P-Q opposite. R right of P → R at P+1. S opposite R. T at end → T at col 1 or 5; identify which by elimination.

Watch out: "Right of P" — depends on which row P is in (P faces S → R is to P's right means R is at P+1 to P's right, but visually P is in row 1 facing south so to spectator R is to P's left). Convention: stick to "clockwise from P".

P7. "Diagonally opposite" — different row + different column index.

P8. "Person opposite to X is to immediate left of Y" — find X's column, then Y is at column +1 or −1.

P9. When 3 people in row 1 face N + 2 face S (mixed), each has its own left/right.

P10. Row-1 clue + Row-2 clue interlocking — solve column by column.

G.3 CIRCULAR (8 people, all facing centre) (5)

P11. Eight A-H around circle facing centre. A is 3rd to left of B. C is between A and D. E is opposite F. G is to immediate right of E. H sits between F and B. Find arrangement.

Method. Position 1 (top), going clockwise 1-8. "Left" facing centre = clockwise direction (the direction the person's left hand points). Place A and B with A 3rd-to-left-of-B → if B at 1, A at 4 (going CW = left). Use further clues.

Watch out: "Left of A" facing centre is OPPOSITE to "left of A" facing outward. Always note facing.

P12. Mixed-facing circular: 4 face centre, 4 face outward. "Right of X" flips by facing.

P13. Specific clue: "Q is 4th from P" — in 8-circle, 4th away = directly opposite.

P14. "Three people sit between A and B" (in 8-circle going one way) — A and B are at positions diff 4 apart.

P15. "Neither X nor Y is adjacent to Z" — Z's two neighbours are not X, not Y.

G.4 FLOOR + ATTRIBUTE puzzles (5)

P16. 7 people on 7 floors (1=ground, 7=top), each likes a different colour (Red, Blue, Green, Yellow, Orange, Pink, Black). A on 5th floor likes Red. B is 2 floors above C, who likes Blue. D likes Green and lives on a floor below A. E lives on top floor and doesn't like Yellow or Orange. F lives between A and the colour-Pink person. G likes Black.

Method. Build a 2-column table: Floor + Person + Colour. Lock A=5, Red. B above C by 2. Try C floors: C=1→B=3; C=2→B=4; C=3→B=5 (clash); C=4→B=6. Test against other clues.

Watch out: "Below A" means floors 1-4; A above means 6-7.

P17. Floor + flat-number + brand puzzle (3-attribute) — make 3 sub-tables.

P18. Floor + age — combine ranking with floor positions.

P19. "Even floor" + "odd-numbered flat" sub-clues — narrow possibilities aggressively.

P20. "Number of floors between A and B is same as between C and D" — equation in floor numbers.

G.5 SCHEDULING — day / month / date (5)

P21. 7 friends meet on Mon-Sun. P meets on Wed. Q meets 2 days after P. R meets day after Q. S meets day before P. T meets between R and the meeting on Sun. U on Mon. Find day for each.

Method. P=Wed. Q=Fri (Wed+2). R=Sat. S=Tue. U=Mon. T between R (Sat) and Sun → T at... actually T must be between Sat and Sun (no day between!) — re-examine. T meets between R (Sat) and the Sun meeting (someone on Sun). So someone (not Q, R, P, S, U, T) meets Sun → that's the 7th person = the remaining V. T at? Re-read.

Lesson. Keep clue precision in mind; "between" sometimes implies strict (excluding endpoints) or loose (inclusive).

P22. Month puzzle: 8 people in 8 months (Jan-Aug). Each in different month. Clue cluster gives unique solution.

P23. Date + month combo: 5 people have birthdays, year given. Use day-of-week calculation if needed.

P24. "Same day in different weeks" — modulo arithmetic.

P25. "Two consecutive months" — pair of clues constraints adjacent positions.

G.6 INPUT-OUTPUT machine (3)

P26. Input: `12 24 6 18 9`. Step 1: `6 12 18 24 9`. Find rule.

Method. Compare element-by-element. - 6 = smallest; was at position 3, now at position 1. - 12, 18, 24 = sorted ascending; were at positions 1, 4, 2. - 9 stays at position 5.

So rule = "Move elements ≤ 12 to front in ascending order; rest stay". Or "ascending sort of left half". Test rule on Step 2.

P27. Word + number machine: words sorted alphabetically, numbers by digit-sum.

P28. Index-shift: each element shifts by its index position.

G.7 DATA SUFFICIENCY — 3 statement variant (2)

P29. Q: Is x even? (I) x is divisible by 4. (II) x is divisible by 6. (III) x is positive integer.

Method. (I) alone: divisible by 4 → x even ✓. **Sufficient.** (II) alone: divisible by 6 → x = 6k → even ✓. **Sufficient.** (III) alone: positive int — could be even or odd. Insufficient. Best answer: I or II alone suffices; III alone doesn't.

P30. Q: Find Mr X's age. (I) X is older than Y by 5 yrs. (II) Y is 30 yrs old. (III) X + Y = 65.

Method. (II) + (I) → X = 35 (sufficient). (III) + (I) → 2Y + 5 = 65 → Y = 30 → X = 35 (sufficient). (II) + (III) → Y = 30 → X = 35 (sufficient). Any TWO suffice. Best answer: any two of I, II, III.

PART H — ADVANCED LOGICAL REASONING (worked examples)

H.1 Statement & Course of Action (3)

Q1. Statement: "Petrol prices have risen by 30% in the last 6 months." Course of action: (I) Govt should regulate petrol prices. (II) Increase domestic production. Which follows?

Method. Test if action directly addresses statement. (I) regulation is one direct response. (II) increase production is also direct. Both follow.

Q2. Statement: "Many students are failing in math." CoA: (I) Mandatory remedial classes. (II) Cancel math from syllabus. (II) is extreme; reject. (I) follows.

Q3. Statement: "Mumbai air quality has deteriorated to AQI 350." CoA: (I) Restrict vehicles in city. (II) Ban diesel sales nationwide. (I) addresses cause; (II) is extreme + national-level vs local issue. Only I follows.

H.2 Statement & Conclusion (3)

Q4. Statement: "All students who studied for 8+ hours daily passed the exam." Conclusions: (I) Studying longer guarantees passing. (II) Some students passed without 8 hrs study.

Method. (I) is "if and only if" — statement says "8+ hr → passed", not the converse. (I) doesn't follow. (II) statement doesn't say anything about <8 hr students. (II) doesn't follow strictly. Neither follows.

Q5. Statement: "All ministers attended the meeting. Some ministers spoke at the meeting." Conclusions: (I) Some who attended spoke. (II) All who spoke are ministers.

Method. (I) follows from "some ministers spoke" ⊂ "all attended". (II) follows from "ministers spoke" ⊂ "all who spoke" — only if no non-minister also spoke (statement doesn't say). So (II) doesn't strictly follow. Only (I).

Q6. Statement: "Smoking causes cancer." Conclusion: (I) Smoking is harmful. (II) Non-smokers don't get cancer.

Method. (I) directly follows. (II) is a converse, doesn't follow (cancer has many causes). Only (I).

H.3 Statement & Argument (3)

Q7. Should reservation in education be abolished? - Argument 1 (Yes): "Equality of opportunity is a constitutional principle." - Argument 2 (No): "Centuries of injustice need correction; affirmative action provides it."

Method. Both are STRONG (relevant to question; substantial). The answer is "both arguments are strong".

Q8. Should children be given homework? - Yes: "Homework reinforces learning." - No: "Children also need play time."

Both arguments are strong + valid; both follow.

Q9. Should the voting age be lowered to 16? - Yes: "16-year-olds are aware of issues today." - No: "16 is too young for political maturity."

Both have merit; both follow.

H.4 Cause & Effect (3)

Q10. Statement A: "Inflation rose to 8%." Statement B: "RBI raised interest rates." - A could be cause; B could be effect (RBI responds to inflation by raising rates). - B → A is also possible (rates affect inflation with lag). - Most likely: A is cause, B is effect (in the immediate term).

Q11. A: "Heavy rain forecast." B: "Schools declared closed." - A → B (schools close due to rain forecast).

Q12. A: "PM announced new healthcare scheme." B: "Stock market rose." - Loose correlation, no clear causation. "Independent events".

PART I — MASTER REVISION CHECKLIST (1-page exam-day)

Compass + Family tree + Cube + Alphabet rows ready in head.

Type-recognition cues: - "sit in a row / circle" → Seating arrangement → draw skeleton, lock most-constrained - "lives on floor" → Floor puzzle → draw n-floor stack - "P + Q means" → Coded relations → decode key first - "@ # & $ *" → Coded inequality → decode → chain scan - "All / Some / No" → Syllogism → Venn diagrams - "Step 1, Step 2" → Input-output → input vs Step 1 first - "I and II, I or II" → Data sufficiency → test each alone - "Pointing to photo" → Blood relations → identify speaker - "Walks 5 km north" → Direction → N-E cross - "Mirror image" → vertical-axis flip + reverse order

Cube formula ($n \times n \times n$ cube painted then cut to 1 cm^3): - Corners: 8 (3-painted) - Edges: $12(n-2)$ (2-painted) - Faces: $6(n-2)^2$ (1-painted) - Interior: $(n-2)^3$ (0-painted)

Inequality chain breaks at sign reversal. > beats ≥ beats =.

Syllogism: Conversion of "Some X are Y" → "Some Y are X" ✓.

Banks priority: puzzles > seating > syllogism > coded inequality > DS > input-output.

SSC priority: series > analogy > classification > coding > non-verbal.

> *Print this 1-page card. Revise the morning before exam. THIS gets you to 95%+ on Reasoning.*

Drill all 50 + the mini-mock thrice for >90% mastery. Then attempt 4-5 PYQ papers from the past 3 years to lock in real exam-pace.